Design, Construction, Operation, Maintenance, and Inspection of Terminal & Tank Facilities

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STEP
Strategies for Today's Environmental Partnership

American Petroleum Institute
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Washington, D.C. 20005

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STEP

One of the most significant long-term trends affecting the future vitality of the petroleum industry is the public's concerns about the environment. Recognizing this trend, API member companies have developed a positive, forward looking strategy called STEP: Strategies for Today's Environmental Partnership. This program aims to address public concerns by improving industry's environmental, health and safety performance; documenting performance improvements; and communicating them to the public. The foundation of STEP is the API Environmental Mission and Guiding Environmental Principles. API standards, by promoting the use of sound engineering and operational practices, are an important means of implementing API's STEP program.

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- To operate our plants and facilities, and to handle our raw materials and products in a manner that protects the environment, and the safety and health of our employees and the public.
- To make safety, health and environmental considerations a priority in our planning, and our development of new products and processes.
- To advise promptly appropriate officials, employees, customers and the public of information on significant industry-related safety, health and environmental hazards, and to recommend protective measures.
- To counsel customers, transporters and others in the safe use, transportation and disposal of our raw materials, products and waste materials.
- To economically develop and produce natural resources and to conserve those resources by using energy efficiently.
- To extend knowledge by conducting or supporting research on the safety, health and environmental effects of our raw materials, products, processes and waste materials.
- To commit to reduce overall emissions and waste generation.
- To work with others to resolve problems created by handling and disposal of hazardous substances from our operations.
- To participate with government and others in creating responsible laws, regulations and standards to safeguard the community, workplace and environment.
- To promote these principles and practices by sharing experiences and offering assistance to others who produce, handle, use, transport or dispose of similar raw materials, petroleum products and wastes.

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Design, Construction, Operation, Maintenance, and Inspection of Terminal & Tank Facilities

Manufacturing, Distribution, and Marketing Department

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FOREWORD

This standard was developed to guide the management of terminals and tanks in a manner that protects the environment and the safety of workers and the public. This standard is intended for petroleum terminal and tank facilities associated with marketing, refining, pipeline, and other similar facilities. This standard may be used as a resource and management guide by those responsible for such facilities and by those working on their behalf. This standard is a compilation of industry knowledge, information, and management practices for all relevant aspects of terminal and tank operations aggregated into an overview document comprising best practices. In instances where greater detail or additional information may be helpful or needed, this standard references other API publications or similar industry guides and standards. It is intended to be consistent with, but is not a substitute for, any applicable local, state, or federal regulations.

Some provisions in this standard, as indicated by the use of the word *shall*, are mandatory and have to be followed to meet the intent of this standard. Some provisions are recommended, as denoted by the word *should*, but are not mandatory. These provisions will need to be considered based on site-specific factors. Still other provisions are optional, as denoted by the word *may*. Typically, these will be given where a range of good options exists.

To foster greater awareness and assist the industry in addressing environmental, health, and safety concerns, the API has undertaken the development of this single document aggregating the various standards, specifications, and recommended practices on the design, construction, operation, inspection, and maintenance of petroleum terminals and tanks. API also has significant research underway to assist members in addressing issues of groundwater protection and remediation of soil contamination. This research includes the evaluation of improved leak detection technology and the evaluation of better methods to detect and remediate groundwater and soil contamination.

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Design, Construction, Operation, Maintenance, and Inspection of Terminal & Tank Facilities

SECTION 1—GENERAL

1.1 Scope and Purpose

This standard covers the design, construction, operation, inspection, and maintenance of petroleum terminal and tank facilities associated with marketing, refining, pipeline, and other similar activities as stipulated in 1.1.1 through 1.1.7. This standard covers the issues of site selection and spacing, pollution prevention and waste management, safe operation, fire prevention and protection, tanks, dikes and berms, mechanical systems, product transfer, corrosion protection, structures, utilities and yard, and removals and decommissioning.

The purpose of this standard is to aggregate a wide base of current industry experience, knowledge, information, and management practices into a cohesive standard comprising a range of best practices.

Information is presented in this standard in the form of recommendations (designated by the use of the word *should*) and in the form of mandates (designated by the use of the word *shall*) (refer to 1.4.13 and 1.4.14). This standard incorporates by reference a number of other standards and recommended practices. The distinctions between mandatory, recommended, and optional provisions in the referenced documents are not changed by the nature of their reference in this standard.

The values stated for this standard are in U.S. customary units with the International System of units (SI) provided in parentheses.

The petroleum industry is engaged in the manufacture, storage, transportation, blending, and distribution of crude oil and refined products. Individual terminal facilities and plants may perform one or more of these functions. These facilities represent diverse operations ranging from small distribution facilities (for example, bulk plants), to large storage and distribution terminals (for example, wholesale plants), and up to large integrated facilities (for example, refining, grease production, and oil blending). The specific application of this standard within those various types of operations is itemized in 1.1.1 through 1.1.7.

1.1.1 PETROLEUM TERMINALS

Petroleum terminals consist of tank farms, loading and unloading areas, pipeline manifolds, storage areas, warehouses, docks, garages, laboratories, and office buildings. Products may be received and distributed by pipeline, marine transport, rail, or truck. Bulk quantities of refined products are stored in aboveground tanks for distribution in smaller quantities to industrial customers, to commercial consumers, and to retail and wholesale marketing facilities. Petroleum terminals may also store and distribute petroleum products in package and container quantities.

1.1.2 PIPELINE TANKAGE FACILITIES

Pipeline tankage facilities consist of tanks and tank farms used to do the following: (a) provide surge relief from pipeline operations, and (b) receive petroleum (crude oil and refined products) from pipelines, trucks, railcars, or marine vessels for transport by pipeline to other facilities, such as petroleum terminals, refineries, other pipeline tankage, and marine facilities (see 49 CFR 195).

1.1.3 REFINERY FACILITIES

Loading and unloading areas, docks, and some refinery tankage facilities are included in this standard. Refinery tankage covered by this standard excludes those aboveground tanks or groups of tanks as defined in 1.4.10 (for example, process tanks).

Examples of applicable covered refinery tankage include tanks that are used to accomplish the following:

a. Receive incoming crude oil.
b. Store intermediate products or components outside of the refinery process units.
c. Store finished products for shipment by truck, marine transport, rail, or pipeline.

Examples of refinery tankage and other equipment specifically excluded are as follows:

a. Those that are integral to refinery process equipment.
b. Refinery tanks and other equipment located within the battery limits of process units.
c. Process equipment located outside the process unit battery limits.
d. Tanks that contain materials, such as additives, used in refinery processes or utility systems.

1.1.4 BULK PLANTS

Although bulk plants typically handle smaller quantities of product, operations and facilities at these plants are similar to those at petroleum terminals. Bulk plants typically receive and distribute product by truck, although some are serviced by rail, marine transport, or pipeline. Bulk plants may also store an inventory of petroleum products in container quantities.

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1.1.5 LUBE BLENDING AND PACKAGING FACILITIES

Lube oil blending and packaging facilities blend refined base stock products with additives and then package the finished products in drums, pails, portable tanks, or consumer-size containers or ship to consumers in bulk. The additives and lube base stocks may be received and stored either in bulk or in containers. Lube blending and packaging facilities typically include warehouses, blending and packaging areas, quality control lab, base stock and additive storage area, shipping and receiving areas, and office buildings.

1.1.6 ASPHALT PLANTS

Asphalt plants receive asphalt from petroleum refineries and blend it with additives to produce paving, roofing, and industrial grade asphalt products. Asphalt facilities typically consist of a laboratory for quality control, a rail siding or ship dock, an aboveground tank farm, a warehouse, one or more unloading areas for raw materials and products, a manufacturing area, a package heating system, a truck scale, a loading rack, and an office.

1.1.7 AVIATION SERVICE FACILITIES

Aviation service facilities store light petroleum fuels in aboveground or underground storage tanks. Services provided may include the following: refueling, defueling, de-icing, washing, maintenance, and repair of aircraft. Aircraft fuel may be loaded into refueling trucks that service the aircraft, or dispensed directly into aircraft from a fixed dispenser system or hydrant system cart.

1.2 Nonapplicability and Retroactivity

This standard does not apply to the following installations:

a. Those covered by API Standard 2510, *Design and Construction of Liquified Petroleum Gas (LPG) Installations*, and API Recommended Practice 12R1, *Recommended Practice for Setting, Maintenance, Inspection, Operation, and Repair of Tanks in Production Service*.
b. Retail facilities, such as service stations.
c. Refinery process equipment, refinery process tanks located within the battery limits of the process units, and gas processing equipment (see 1.1.3).
d. Tanks or vessels less than 1100 gallons (4164 liters) in storage capacity.
e. Tanks that are part of oil and gas production, natural gas processing plants, or offshore operations.
f. Production, pipeline, and truck crude oil gathering facilities.

The design and construction provisions of this standard are intended for application at new facilities or installations. Application of the design and construction provisions of this standard to facilities, equipment, structures, or installations that are already in place, that are in the process of construction, or that are installed before the date of this publication should be evaluated when circumstances merit. Such an evaluation should consider the individual circumstances, existing conditions at the installation, and overall benefit of applying the design and construction provisions.

The operation, inspection, and maintenance provisions in various sections of this standard are intended for application to both new and existing facilities or installations.

1.3 Referenced Publications

The most recent edition or revision of the following publications may provide additional information for users of Standard 2610. The provisions of these publications are incorporated into Standard 2610 only as explicitly specified in the text.

AAR[1]
Mechanical Division Standards

ABS[2]
Rules for Building and Classing Steel Vessels

ACI[3]
318 *Building Code Requirements for Reinforced Concrete*

AISC[4]
S310 *Specifications for the Design, Fabrication, and Erection of Structural Steel Buildings*

API
Spec 5L *Specification for Line Pipe*
Spec 6FA *Specification for Fire Test for Valves*
Spec 12P *Specification for Fiberglass Reinforced Plastic Tanks*
RP 12R1 *Recommended Practice for Setting, Maintenance, Inspection, Operation, and Repair of Tanks in Production Service*
Publ 306 *An Engineering Assessment of Volumetric Methods of Leak Detection in Aboveground Storage Tanks*
Publ 307 *An Engineering Assessment of Acoustic Methods of Leak Detection in Aboveground Storage Tanks*
RP 500 *Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities*

[1]Association of American Railroads, 59 East Van Buren Street, Chicago, Illinois, 60605.
[2]American Bureau of Shipping, 65 Broadway, New York, New York, 10006.
[3]American Concrete Institute, P.O. Box 19150, Detroit, Michigan 48219.
[4]American Institute of Steel Construction, 1221 Avenue of the Americas, New York, New York 10020.

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RP 540 *Electrical Installations in Petroleum Processing Plants*
Std 570 *Inspection, Repair, Alteration, and Rerating of In-Service Piping Systems*
Std 607 *Fire Test for Soft-Seated Quarter Turn Valves*
Std 610 *Centrifugal Pumps for General Refinery Service*
Std 620 *Design and Construction of Large Welded, Low-Pressure Storage Tanks*
Std 650 *Welded Steel Tanks for Oil Storage*
RP 651 *Cathodic Protection of Aboveground Petroleum Storage Tanks*
RP 652 *Lining of Aboveground Petroleum Storage Tank Bottoms*
Std 653 *Tank Inspection, Repair, Alteration, and Reconstruction*
Std 674 *Positive Displacement Pumps - Reciprocating*
Std 675 *Positive Displacement Pumps - Controlled Volume*
Std 676 *Positive Displacement Pumps - Rotary*
RP 1004 *Bottom Loading and Vapor Recovery for MC-306 Tank Motor Vehicles*
RP 1110 *Pressure Testing of Liquid Petroleum Pipelines*
RP 1124 *Ship, Barge, and Terminal Hydrocarbon Vapor Collection Manifolds*
RP 1125 *Overfill Control Systems for Tank Barges*
Publ 1500 *Storage and Handling of Aviation Fuels at Airports*
Publ 1529 *Aviation Fueling Hose*
Publ 1581 *Specifications and Qualification Procedures for Aviation Test Fuel Filter/Separators*
RP 1604 *Removal and Disposal of Used Underground Petroleum Storage Tanks*
RP 1615 *Installation of Underground Petroleum Storage Systems*
Publ 1621 *Bulk Liquid Stock Control at Retail Outlets*
RP 1626 *Storing and Handling Ethanol and Gasoline-Ethanol Blends at Distribution Terminals and Service Stations*
RP 1627 *Storage and Handling of Gasoline-Methanol/Cosolvent Blends at Distribution Terminals and Service Stations*
Publ 1628 *A Guide to the Assessment and Remediation of Underground Petroleum Releases*
Publ 1629 *Guide for Assessing and Remediating Hydrocarbons in Soil*
RP 1631 *Interior Lining of Underground Storage Tanks*
RP 1632 *Cathodic Protection of Underground Petroleum Storage Tanks and Piping Systems*
Publ 1637 *Using the API Color Symbol System to Mark Equipment and Vehicles for Product Identification at Service Stations and Distribution Terminals*
Publ 1638 *Waste Management Practices for Petroleum Marketing Facilities*
Std 2000 *Venting Atmospheric and Low-Pressure Storage Tanks*
RP 2003 *Protection Against Ignitions Arising Out of Static, Lightning, and Stray Currents*
Publ 2009 *Safe Welding and Cutting Practices in Refineries, Gasoline Plants, and Petrochemical Plants*
Std 2015 *Safe Entry and Cleaning of Petroleum Storage Tanks*
Publ 2021 *Fighting Fires in and Around Flammable and Combustible Liquid Atmospheric Storage Tanks*
Publ 2026 *Safe Descent onto Floating Roof Tanks*
Publ 2202 *Dismantling and Disposing of Steel from Aboveground Leaded Gasoline Storage Tanks*
RP 2220 *Improving Owner and Contractor Safety Performance*
RP 2350 *Overfill Protection for Petroleum Storage Tanks*
STD 2510 *Design and Construction of Liquified Petroleum Gas (LPG) Installations*
Publ 2517 *Evaporative Loss From External Floating-Roof Tanks*
Publ 2518 *Manual of Petroleum Measurement Standards*, Chapter 19, "Evaporative Loss Measurement," Section 1
Publ 2519 *Evaporation Loss from Internal Floating-Roof Tanks*
Std 2550 *Method for Measurement and Calibration of Upright Cylindrical Tanks*
Publ 2557 *Vapor Collection and Control Options for Storage and Transfer Operations in the Petroleum Industry*
Publ 4588 *Development of Fugitive Emission Factors and Emission Profiles for Petroleum Marketing Terminals*
Publ 4602 *Minimization, Handling, Treatment, and Disposal of Petroleum Terminal Waste Waters*

Manual of Petroleum Measurements Standards

AREA[5]

Manual for Railway Engineering, Volume 1 & 2

[5]American Railway Engineering Association, 50 F Street N.W., Washington, D.C. 20005.

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ASM International[6]

ASM Metals Handbook, Volume 13, *Corrosion*

ASME[7]

B16.5 *Pipe Flanges and Flanged Fittings*
B16.9 *Factory Made Wrought Steel Buttweld Fittings*
B16.11 *Forged Steel Fittings, Socket Welding and Threaded*
B16.20 *Ring Joint Gaskets and Grooves for Steel Pipe Flanges*
B16.21 *Non-metallic Flat Gaskets for Pipe Flanges*
B16.28 *Wrought-Steel Buttwelding Short-Radius Elbows and Returns*
B16.47 *Large Diameter Steel Flanges*
B31.3 *Chemical Plant and Petroleum Refinery Piping*
B31.4 *Liquid Petroleum Transportation Piping Systems*
B31.5 *Refrigeration Piping Systems*
B73.1M *Specifications for Horizontal End Suction Centrifugal Pumps for Chemical Process*

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A194 *Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service*
A325 *Specification for High Strength Bolts for Structural Steel Joints*

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D-1.1 *Welding in Building Construction*

AWWA[10]

B300 *Disinfection Standards*
C100 *Iron Pipe Standards*

BOCA[11]

BOCA National Building Code
BOCA National Fire Code

CFR[12]

28 CFR 36 *Nondiscrimination on the Basis of Disability by Public Accommodations and in Commercial Facilities*
29 CFR 1910 *Occupational Safety and Health Standards for Industry*
29 CFR 1926 *Occupational Safety and Health Standards for Construction*
33 CFR 26 *Vessel Bridge to Bridge Radio Telephone Regulation*
33 CFR 126 *Handling of Explosives or other Dangerous Cargoes Within or Contiguous to Waterfront Facilities*
33 CFR 154 *Coast Guard Regulations for Facilities Transferring Oil or Hazardous Material in Bulk*
33 CFR 155 *Coast Guard Oil or Hazardous Material Pollution Prevention Regulations for Vessels*
33 CFR 156 *Coast Guard Regulations on Oil and Hazardous Material Transfer Operations*
40 CFR 51 *EPA Regulations on Preparation of Implementation Plans*
40 CFR 52 *EPA Regulations on Approval and Promulgation of Implementation Plans*
40 CFR 60 *EPA Regulations on Standards of Performance for New Stationary Sources*
40 CFR 61 *EPA Regulations on National Emission Standards for Hazardous Air Pollutants*
40 CFR 112 *EPA Regulations on Oil Pollution Prevention*
40 CFR 122 *EPA National Pollutant Discharge Elimination System Permit Regulations*
40 CFR 123 *EPA Regulations on State NPDES Permit Program Requirements*
40 CFR 125 *EPA National Pollutant Discharge Elimination System Regulations*
40 CFR 262 *EPA Regulations for Hazardous Waste Generators*
40 CFR 264 *EPA Regulations for Owners and Operators of Permitted Hazardous Waste Facilities*
40 CFR 265 *EPA Interim Status Standards for Owners and Operators of Hazardous Waste Facilities*
40 CFR 280 *EPA Technical Standards and Corrective Action Requirements for Owners and Operators of Underground Storage Tanks*
40 CFR 311 *EPA Worker Protection Regulations*
40 CFR 403 *EPA Pretreatment Standard*
46 CFR 39 *Vapor Control Systems*
49 CFR 195 *Transportation of Hazardous Liquids by Pipeline*

[6]ASM International, Materials Park, Ohio 44073.
[7]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[8]American Society of Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.
[9]American Welding Society, 550 LeJuene Road NW, Miami, Florida, 33135.
[10]American Water Works Association, 6666 West Quincy Avenue, Denver, Colorado 80235.
[11]Building Officials & Code Administrators, 4051 W. Flossmoor Road, Country Club Hills, Illinois 60478.
[12]Code of Federal Regulations, Government Printing Office, North Capital Street, Washington, D.C. 20402.

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ICOS/OCIMF/IAPH[15]

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IES[16]

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SP-75 *1993 Specification for High Test Wrought Butt Welding Fittings*
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RP0169-92 *Control of External Corrosion on Underground or Submerged Metallic Piping Systems*
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Std TM-01-70 *Visual Standard*

NFPA[19]

10 *Portable Fire Extinguisher*
11 *Low Expansion Foam and Combined Agent Systems*
15 *Water Spray Fixed Systems*
16 *Installation of Deluge Foam-Water Sprinkler Systems and Foam Water Spray Systems*
20 *Installation of Centrifugal Fire Pumps*
22 *Water Tanks for Private Fire Protection*
24 *Installation of Private Fire Service Mains and Their Appurtenances*
30 *Flammable and Combustible Liquids Code*
69 *Explosion Prevention Systems*
70 *National Electrical Code*
77 *Static Electricity*
101 *Safety to Life from Fire in Buildings and Structures*
111 *Stored Electrical Energy Emergency and Standby Power Systems*
496 *Purged and Pressurized Enclosures for Electrical Equipment in Hazardous (Classified) Locations*
497A *Classification of Class I Hazardous Locations for Electrical Installations in Chemical Process Areas*
600 *Industrial Fire Brigades*
780 *Lightning Protection Code*

OCIMF[20]

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SBCC[21]

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Standard Fire Code

SSPC[22]

SP1 *Solvent Cleaning*
SP2 *Hand Tool Cleaning*
SP3 *Power Tool Cleaning*
SP5 *White Metal Blast Cleaning*
SP6 *Commercial Blast Cleaning*
SP7 *Brush-Off Blast Cleaning*
SP10 *Near-White Metal Blast Cleaning*
Steel Structures Painting Manual, Volume 1, *Good Painting Practice*
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[13]United States Environmental Protection Agency, Research Triangle Park, NC 27711.
[14]International Conference of Building Officials, 5360 South Workman Mill Road, Whittier, California 90601.
[15]International Chamber of Shipping, Oil Companies International Marine Forum, and International Association of Ports and Harbors; OCIMF, 6th Floor, Portland House, Stay Place, London, SW1E-5BH.
[16]Illuminating Engineering Society of North America, 345 East 47th Street, New York, New York 10017.
[17]Manufacturers Standardization Society of the Valves and Fittings Industry, 127 Park St. NE, Vienna, Virginia 22180.
[18]National Association of Corrosion Engineers International, P.O. Box 218340, Houston, Texas 77218-8340
[19]National Fire Protection Association, Battery March Park, Quincy, Massachusetts 02269.
[20]Oil Companies International Marine Forum, 6th Floor, Portland House, Stag Place, London, SW1E 5BH.
[21]Southern Building Code Congress, Inc., 900 Montclair Road, Birmingham, Al 35213-1206.
[22]Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, Pennsylvania 15213.
[23]Underwriters Laboratories, 333 Pfingsten Road, Northbrook, Illinois 60062.

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1.4 Definitions

Terms used in this standard are defined in 1.4.1 through 1.4.15.

1.4.1 *Aboveground Tank* as defined within the context of this standard refers to atmospheric vertical or horizontal, cylindrical, closed-top or open-top, steel or fiberglass aboveground storage containers of various sizes and capacities over 1100 gallons (4164 liters) and whose entire bottom is supported uniformly on the ground, completely on saddles, or other supports.

1.4.2 *Berm* refers to the annular area around the tank, inside the dike, normally used for access to the tank and the equipment surrounding it.

1.4.3 *Combustible Liquid* refers to a liquid having a flash point at or above 100°F (37.8°C). See NFPA 30 for discussion of combustible liquid classification.

1.4.4 *Facility* refers to any building, structure, installation, equipment, pipeline, or other physical feature used in petroleum refining, storage, transportation, and distribution. The boundaries of a facility may depend on several site specific factors, including but not limited to, the ownership or operation of buildings, structures, and equipment on the same site, and the types of activity at the site.

1.4.5 *Flammable Liquid* refers to a liquid having a flash point below 100°F (37.8°C) and having a vapor pressure not exceeding 40 pounds per square inch (absolute) (2068 mm Hg) at 100°F (37.8°C) shall be known as Class I liquid (see NFPA 30).

1.4.6 *Installations* refers to tanks, pumps, compressors, accessories, controls, piping, and all other associated equipment required for the receipt, transfer, storage, blending, packaging, and shipment of petroleum products.

1.4.7 *May* indicates provisions that are optional and, consequently, are at the discretion of the designer or operator.

1.4.8 *Must* indicates important legal or safety considerations.

1.4.9 *Petroleum* is any crude oil, liquid, or gaseous complex combination of hydrocarbons and related derivatives (natural or manmade) that may be processed from crude oil for fractions including the following: natural gas, gasoline, naphtha, kerosene, fuel and lubricating oils, paraffin wax, additives, asphalt, and various derivative products.

1.4.10 *Process tank* is an aboveground or underground tank that forms a part of a refining process through which there is a steady, variable, recurring, or intermittent flow of materials during the operation of the process. This definition excludes tanks used for the storage of products or raw materials before their introduction into the refining process or for the storage of finished products or by-products from the refining process.

1.4.11 *Release Prevention Barriers* (RPBs) include steel bottoms, synthetic materials, clay liners, and all other barriers or combination of barriers placed in the bottom of or under an aboveground storage tank that have these functions: (a) preventing the escape of contaminated material, and (b) containing or channeling released material for leak detection. This process is covered in detail in the nonmandatory Appendix I of API Standard 650.

1.4.12 *Release Prevention System* (RPS) refers to the suite of API Standards that is designed to maintain aboveground storage tank integrity, thus protecting the environment. These standards cover such things as the frequency of internal inspections, lining the bottom of the tank interior, fitting the tank with RPBs, installing cathodic protection, or some combination of these measures depending on the operating environment and service of the tank.

1.4.13 *Shall* indicates provisions that are mandatory to meet this API standard.

1.4.14 *Should* indicates provisions that are recommended but not mandatory. Implementation of these provisions will be made based upon consideration of the following, as appropriate: (a) risk/benefit analysis, (b) company standards, (c) company experience, and (d) company philosophy.

1.4.15 *Underground tank* is a buried container that has a capacity of more than 110 gallons (416 liters) and is used to store petroleum, additives, or other products for later use.

1.5 Governmental Requirements and Reviews

Reviews should be conducted on a periodic basis to help ensure that the facility meets applicable federal, state, or local requirements. These reviews should include but not be limited to the following:

a. Spill Prevention Control and Countermeasure (SPCC) plans, 40 CFR 112 and other Environmental Protection Agency (EPA) requirements.
b. Occupational Safety and Health Act (OSHA) requirements.
c. Tank or vessel inspection records, registration, or files.
d. Discharge or remediation permits.
e. Air and water permits.
f. Oil Pollution Act of 1990 (OPA 90) regulations and contingency plans.
g. State and local emergency plans.
h. 49 CFR 195 - Department of Transportation (DOT) Transportation of Hazardous Liquid by Pipeline regulations.
i. Superfund Amendments and Reauthorization Act (SARA) Title III requirements.
j. United States Coast Guard (USCG) requirements, 33 CFR 154.
k. Local or State Fire Marshall Permits.

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SECTION 2—SITE SELECTION AND SPACING REQUIREMENTS

2.1 Overview

This section covers the site selection and spacing requirements for the design and construction of new terminal facilities.

2.2 Site Selection

A site selection criteria should be established that minimizes the potential risk to property adjacent to the terminal facility and the risk presented to the terminal facility by an incident on the adjacent property. The following factors should be considered when selecting a site for a new terminal facility:

a. Proximity to public ways.
b. Proximity to waterways and other surface waters.
c. Risk to and from adjacent facilities.
d. Types and quantities of products to be stored.
e. Proximity to populated areas.
f. Present and predicted development or rezoning of adjacent properties and the terminal property.
g. Topography of the site including elevation, slope, and drainage.
h. Assessment of site environmental conditions including soil and groundwater conditions, such as depth of the water table and type of aquifer. Refer to API Publication 1628 for information on assessment technology.
i. Availability of off-site emergency services and access for emergency response.
j. Location and availability of required utilities including the following: electricity, sewers, potable water, fire water, steam, air, and natural gas.
k. Requirements for receipt and shipment of products including the following: over-the-road, rail, marine, and pipeline facilities.
l. Codes, permits, license requirements, and regulations applicable to construction, operation, and environmental requirements including these: zoning requirements; building permits and codes; operating permits; fire marshall permits; and air, water, and waste discharge permits.
m. Proximity to existing supply and distribution facilities.

2.3 Spacing Requirements

2.3.1 GENERAL

2.3.1.1 The spacing provisions presented in this section are intended to minimize the fire hazard (see 5.2) and the exposure risk to adjacent tanks, equipment, and buildings if a fire or other incident were to occur. The specific spacing distances necessary for safe operations require identification of the hazards involved (see 4.2).

2.3.1.2 Site specific reviews of the potential risks should be used to estimate setback distances needed to limit the exposure to or from adjacent facilities. Sources of information on this subject may be found in NFPA 30 and the ICBO *Uniform Fire Code*.

2.3.2 SPACING OF ABOVEGROUND STORAGE TANKS

2.3.2.1 The provisions for the location of aboveground storage tanks with respect to property lines, public ways, and important buildings on the same property are listed in NFPA 30 or other locally recognized requirements, such as the ICBO *Uniform Fire Code*.

2.3.2.2 The requirements for spacing (shell-to-shell) between any two adjacent aboveground storage tanks are listed in NFPA 30, or other locally recognized requirements, such as the ICBO *Uniform Fire Code*.

2.3.2.3 The impact on spacing requirements by the use of dikes, berms, and drainage swales shall be considered during the initial layout (see Section 7).

2.3.3 SPACING OF TANK TRUCK LOADING RACKS

Tank truck loading racks for flammable and combustible liquids should be located near the terminal access way and should be oriented to provide a one-way traffic pattern for entrance and exit with clear and direct access to the terminal exit. The requirements for the spacing or location of truck loading racks from aboveground tanks, warehouses, and other facility buildings or adjoining property lines are listed in NFPA 30 or other locally recognized requirements, such as the ICBO *Uniform Fire Code*. Refer to 9.3 for additional requirements. Two-way drives are also common to accommodate trucks with manifolds on both sides; however, these locations require larger paved areas.

2.3.4 SPACING OF RAIL TANK CAR LOADING RACKS

Rail tank car loading racks for flammable and combustible liquids should be located so as to avoid road blockage when spotting of rail cars. This precaution will help ensure the passage of fire trucks or other emergency vehicles. The requirements for rail facility design are listed in the AAR *Manual for Railway Engineering*, Volumes 1 & 2 and AAR *Mechanical Division Standards*. Other requirements by local railroads or as stipulated in the ICBO *Uniform Fire Code* may apply. Refer to 9.4 for rail tank car loading and unloading requirements.

2.3.5 SPACING OF MARINE FACILITIES

The requirements for the spacing and location of marine facilities handling flammable and combustible liquids are

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listed in NFPA 30 and 33 CFR 154 or other locally recognized requirements, such as the ICBO *Uniform Fire Code*. Refer to 9.5 for marine loading and unloading requirements.

2.3.6 SPACING OF VAPOR CONTROL EQUIPMENT

Combustion vapor control units are a potential ignition source. Some terminals may use fuel source tanks [for example, Liquid Petroleum Gas (LPG) tanks] in conjunction with combustion vapor control units. Electrical classification in areas where such ignition source equipment is to be located should be in accordance with NFPA 70, NFPA 496, NFPA 497A, and API Recommended Practice 500 or other applicable local requirements. See 5.2.1 and 9.7 for additional requirements.

2.3.7 SPACING OF PUMPS AND OTHER MISCELLANEOUS EQUIPMENT

The requirements for the spacing of pumps for the handling of flammable and combustible liquids and other miscellaneous equipment, such as diesel generators and internal combustion engines are listed in NFPA 30 or other locally recognized equivalent requirements, such as the ICBO *Uniform Fire Code*. Refer to 7.2.8 and 8.5 for additional pump requirements.

2.3.8 SPACING OF BUILDINGS

Location of buildings and spacing between buildings should be based on the type of construction, the intended use, the applicable NFPA fire codes, or other locally recognized requirements, such as the ICBO *Uniform Fire Code*, the BOCA *National Building Code*, the ICBO *Uniform Building Code*, the SBCC *Standard Building Code*, or state/municipal building code. Refer to 11.1 for structure requirements.

2.3.9 SPACING OF ELECTRICAL EQUIPMENT

Spacing and location of electrical equipment should consider the requirements of the area's electrical classification. Recommendations for determining the electrical area classification are given in NFPA 30, NFPA 497A, Article 513 and Article 515 of the *National Electrical Code* (NFPA 70), API Recommended Practice 500, or other applicable state/local requirements. Refer to 11.2.2 for electrical utility requirements.

2.3.10 ALTERNATIVES TO SPACING REQUIREMENTS

If the layouts at some sites are so that the recommended spacing requirements may not be met, consideration should be given to increasing the fire preventive aspects of design, such as improved drainage, equipment isolation, increased fire suppression equipment, or an alternative, such as locating equipment in pressurized enclosures. Examples of such applications would be enclosures for motor control centers, diesel generators, or vapor control units. Further information on this subject may be found in NFPA 496.

SECTION 3—POLLUTION PREVENTION AND WASTE MANAGEMENT

3.1 Applicability

This section addresses the methods of pollution prevention and waste management practices in the design, maintenance, and operation of petroleum terminal and tank facilities.

3.1.1 REGULATORY TRENDS

Federal, state, and local regulations continue to increase pollution prevention and waste minimization requirements for petroleum terminal and tank facilities. Regulatory requirements involve use of the following:

a. Placement of limits on air emissions, water discharges, waste generation, waste handling, and waste disposal.
b. Collection of fees for emissions, discharges, and waste disposal.
c. Identification, recordkeeping, and reporting requirements for sources and amounts of air emissions, water discharges, waste generation, waste handling, and waste disposal. Refer to 1.3 and 1.5 for referenced environmental regulations and governmental requirements.

Use of a waste management hierarchy provides a practical means of pollution prevention. Cost savings may accrue from reduction of waste generated and reduced levels of treatment or disposal.

3.2 Waste Management Hierarchy

A waste management hierarchy serves as a guide to pollution prevention activities. The waste management practices described in this section should be considered in the design, operation, and maintenance of facilities.

The four categories within the hierarchy are prioritized in order of general acceptance: Source Reduction/Waste Elimination, Recycling/Reuse, Treatment, and Disposal. The following sections briefly describe each of the four categories and provide examples where appropriate.

3.2.1 SOURCE REDUCTION/WASTE ELIMINATION

Changes in facility design, operation, and maintenance that reduce, avoid, prevent, or eliminate waste generation are

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regarded as source reduction/waste elimination. *Source reduction/waste elimination* refers to a variety of activities that prevent (waste elimination) or decrease (source reduction) the amount of waste created or the toxicity of waste generated. Source reduction and waste elimination activities include, but are not necessarily limited to, the following:

a. Equipment or technology modifications.
b. Procedure and operations modifications.
c. Substitution of materials.
d. Improved housekeeping, training, and inventory control.

Before initiation of significant facility operation modifications, a management of change review may be required (refer to 4.6). Additionally, the proposed modifications may be subject to regulatory review including permit modifications or new permits.

3.2.2 RECYCLING/REUSE

Resource recovery and other similar practices may be implemented to beneficially reclaim, recycle, reuse, or reprocess materials from a facility.

3.2.3 TREATMENT

Any physical, chemical, biological, or thermal process that reduces the volume or toxicity of waste is regarded as treatment.

3.2.4 DISPOSAL

Discharges of waste into air, water, or onto land may constitute disposal of waste. Solid, liquid, and some gaseous waste disposal must be handled at properly permitted disposal facilities. Refer to 40 CFR 60, 40 CFR 61, 40 CFR 122, 40 CFR 123, and other appropriate federal, state, and local requirements.

3.3 Pollution Prevention

Pollution prevention is a multi-faceted concept designed to reduce or eliminate pollutant discharges to air, water, or land. Pollution prevention includes changes in practices, source reduction, beneficial use, and environmentally sound recycling. The aim of pollution prevention programs is to minimize the generation of waste products, as well as minimize inventory loss to the air, water, and land. It is the responsibility of management to enact appropriate pollution prevention strategies that are protective of human health (see Section 4) and the environment, as appropriate. Factors to recognize during the development of pollution prevention programs include facility design, operation procedures (see 4.3), maintenance (see 9.15), training (see 4.7), recordkeeping, and inventory control (see 9.12). Pollution prevention should be implemented through source reduction or elimination, recycling, or re-use. Additionally, the implementation of Pre-startup Safety and Operational Inspections (see 4.8) may assist in determining the effectiveness of pollution prevention programs. The benefits of effective pollution prevention programs may include the following:

a. Reduced potential exposure to workers and community.
b. Lower cost of waste disposal and treatment.
c. Reduced risk to the environment.
d. Reduced or simplified regulatory reporting and recordkeeping.
e. Lower material and operating costs.
f. Reduced long-term liabilities.
g. Improved image and public relations.

3.3.1 AIR

Air emissions of toxic and volatile organic compounds are subject to regulations promulgated in response to the Clean Air Act (CAA) and its amendments and the regulations of state and local agencies. The EPA report *Gasoline Marketing Industry (Stage I): Background Information for Proposed Standards* provides additional information. Federal air emission control requirements and standards applicable to petroleum terminal and tank facilities are included in the "New Source Performance Standards" (NSPS) a subsection of 40 CFR 60. NSPS apply to all new or substantially rebuilt facilities as defined within the CAA regulation. Emission of hazardous air pollutants may also be regulated under the National Emissions Standards for Hazardous Air Pollutants (NESHAP) of 40 CFR 61. Individual State Implementation Plans (SIP) are authorized per 40 CFR 52 for implementation of the CAA requirements. The control of air emissions at facilities is listed in 3.3.1.1 through 3.3.1.5. Specific methods used to control emissions will be dependent upon site specific conditions including the applicable regulatory requirements.

3.3.1.1 Loading Operations

Air emission control requirements for regulated product transfer operations include vapor collection and processing systems, as specified by various federal, state, and local requirements, such as 40 CFR 60. Refer to 9.3.3 and 9.7 for additional information on loading operations and vapor control.

3.3.1.2 Product Storage—Aboveground Tanks

Air emissions of volatile organic compounds (VOC) from aboveground tanks may be controlled by utilizing floating roof tanks, secondary seals on floating roof tanks, tanks with fixed roofs over floating roofs, vapor recovery systems, or other methods. See 3.3.1.4, 3.4.1, and 6.1.3.3 for additional information on aboveground storage tank vapor emissions. Refer also to 40 CFR 60 (NSPS); 40 CFR 61 (NESHAP); and other federal, state, and local requirements for air emission control of aboveground storage tanks.

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3.3.1.3 Product Storage—Underground Tanks

Air emissions of VOC from underground tanks, such as collection of vapors emitted during filling, are generally regulated by State Implementation Plans (SIP) and local regulations. Refer to 40 CFR 52 and state regulations for applicable controls. Refer to API Recommended Practice 1615 and 6.4 of this standard.

3.3.1.4 Fugitive Emissions

Methods for controlling or reducing fugitive vapor emissions from vents, pumps, valves, flanges, and other connecting points should be considered in the design, operation, and maintenance of facilities. Refer to API Publication 4588 for development of fugitive emission factors and emission profiles for petroleum (marketing) terminals.

3.3.1.5 Asbestos, Lead and Other Regulated Airborne Contaminants

Air emissions of dust as a result of removal of asbestos insulation or lead based paint or the use of other regulated toxic materials should be controlled in accordance with federal, state, and local requirements (see 4.4 and 10.2.5.2).

3.3.2 WATER

Pollution prevention may be accomplished by reducing the sources and volume of petroleum contact waste water and the concentrations and toxicity of petroleum compounds in the contact waste water. These reductions are addressed in 3.3.2.1 and 3.3.2.2.

API Publication 4602 provides a range of options for waste water management and treatment at petroleum (marketing) terminals.

3.3.2.1 Petroleum Contact Water

Effective spill containment design and operating procedures reduce the volume of petroleum contact water that must be treated and discharged or disposed. For example, storm water may be segregated from sources of contamination to prevent petroleum contact, thereby, reducing the volume of waste water that must be managed. Refer to 6.2, 8.5.3, 9.3.1, 9.3.2, 9.4.1, 9.4.2, 9.14, 11.3.4, and 11.3.5 for operating procedures and design considerations for waste water management. Refer to API Publication 4602 for design considerations at petroleum (marketing) terminals. SPCC Plan requirements (40 CFR 112) for regulated facilities provide additional guidance. For facilities with marine docks, refer to 33 CFR 126, 33 CFR 154–156, and 40 CFR 311 that regulate transfer of petroleum products across navigable waterways. Refer also to 6.1.7 regarding tank water draw-off.

3.3.2.2 Waste Water Management

Waste water may require treatment to reduce pollutants before discharge, under a National Pollutant Discharge Elimination System (NPDES) permit issued under Federal Regulations 40 CFR 122 and 40 CFR 125 or State permits (SPDES) issued under the authority of 40 CFR 123. Some waste water discharge permits for petroleum terminals may impose discharge limits on specific compounds and impose toxic monitoring or limits on toxic parameters for waste water discharge.

Discharge to a Publicly Owned Treatment Works (POTW) may require pretreatment or other conditions to achieve POTW permit standards before discharge. Refer to 40 CFR 403 for additional information on discharges to POTWs.

Additional opportunities for reduction of waste water pollutants may become evident upon further evaluation of the discharge. Furthermore, toxicity based permits may require further reduction of waste water constituents through pretreatment or post-treatment.

3.3.2.3 Facility Design

Additional facility design guidance is provided for petroleum terminals in API Publication 4602. This API Publication addresses tank farm drainage, tank bottom water, ballast water, storm water, process and containment water, waste water, and test water (refer also to 7.1.3, 8.3.7, and 11.3). Where appropriate, waste water treatment may include:

a. Gravity Separator.
b. Carbon.
c. Air Stripping.
d. Biological Treatment.
e. UV/Oxidation.
f. Dissolved Air Floatation.

3.3.3 SOLID WASTE MANAGEMENT

Solid waste reduction or elimination may be accomplished through facility design, product substitution, maintenance, and operation.

3.3.3.1 Waste Minimization Certification

All hazardous waste manifests require certification that a program is in place to reduce the volume or toxicity of waste generated to the degree economically practical (per 40 CFR 262, 40 CFR 264 and 40 CFR 265). Implementation of pollution prevention designs, maintenance, and operating practices will support this certification.

3.3.3.2 Facility Operation

Facility operation guidance to minimize solid waste in petroleum (marketing) terminals is suggested in API Publi-

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cation 1638. Additionally, the stipulations of Dikes and Berms (Section 7), Corrosion Control (Section 10), Release Prevention (Section 6.1.3.1), and proper handling of recovered petroleum product support waste reduction.

3.4 Waste Management Practices

After consideration of pollution prevention practices, each facility should develop waste management practices that serve to minimize the amount of waste generated on site from normal routine facility operation and maintenance. API Publication 1638 provides guidance and additional information for waste management in support of pollution prevention for petroleum terminals. The waste streams addressed in this standard are grouped into four categories:

a. Waste produced by handling and storage facilities.
b. Waste produced by waste treatment facilities.
c. Vehicular waste.
d. Miscellaneous waste.

Waste management options for wastes identified within each of the above-listed categories are covered in API Publication 1638.

Refer to 3.4.1 through 3.4.3 for suggested waste management practices that are applicable for the facilities covered in this standard.

3.4.1 PRODUCT STORAGE

Source reduction opportunities exist in the design, modification, maintenance, and operation of tanks. Examples of source reduction opportunities are described in 3.4.1.1 and 3.4.1.2. Refer also to Section 6.1.3.

3.4.1.1 Covered Floating-Roof Tanks

The use of a fixed roof over a floating roof will reduce any accumulation of snow or water on the floating roof. Fixed roofs over floating roofs may reduce air emissions depending on the base condition, such as the type and condition of the floating roof and seals (see 6.1.3.3).

3.4.1.2 Tank Cleaning

Improved tank cleaning procedures may reduce the volume of solid waste generated during tank cleaning operations. These procedures include:

a. Extracting all recoverable product for re-use.
b. Use of high pressure/low volume water wash methods.
c. In-situ sludge reduction/reclamation systems or procedures.
d. Recycling of wash water or recycling of other cleaning materials.
e. Extraction of liquids from the products of tank cleaning operations before disposal of solids.

Tank design features, such as sloped bottom design, may reduce the amount of cleaning required as well as the sludge volume generated.

3.4.2 WASTE WATER MANAGEMENT

The design, maintenance, and operation of waste water collection and treatment systems offer opportunities for waste minimization. Several options for waste water management are given in 3.4.2.1 through 3.4.2.3. Additional considerations for petroleum terminals are presented in API Publication 1638 and API Publication 4602. Refer also to 6.1.7, 6.1.8, 9.3.1, 9.3.2, 9.4.1, 9.4.2, and 11.3.

3.4.2.1 Hydrostatic Test Water

Cleaning tanks before performing hydrostatic tests may reduce the generation of contaminated waste water. When disposal of the test water is necessary, its handling should be reviewed with the authorities having jurisdiction before placement of test water in the tank. Unnecessary hydrostatic testing should be avoided as it may produce large volumes of potentially contaminated water that may have to be treated or disposed. Specific exemptions for hydrostatic testing for in-service tanks are given in API Standard 653.

3.4.2.2 Spill Containment

Properly designed spill containment systems at product transfer and storage areas will minimize the volume of storm water that becomes commingled with contaminants within the containment system. A properly designed spill containment system should enclose the minimum area required for containment. A roof or canopy will also minimize the precipitation entering the containment area (refer to 6.1.3, 8.5, 9.3.1, 9.4.1, 9.5.1, 9.6.1, 9.14, 11.3, Section 7, and NFPA 30).

3.4.2.3 Oil/Water Separators

The installation of sand/grit traps or protective mud sumps installed at the separator inlet or at source points, such as catch basins or manholes, will reduce the volume of petroleum contact sediments generated in the separator.

3.4.3 STORM WATER

Facility designs that segregate uncontaminated storm water from petroleum contact water through the use of curbing, hard-piped systems, and other methods will provide source reduction (see 7.1.3 and 11.3.4).

3.4.4 VAPOR RECOVERY UNIT CARBON

Regeneration of used activated carbon from vapor-recovery units is an effective waste reduction technique when the products of regeneration are properly recovered and treated.

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SECTION 4—SAFE OPERATIONS OF TERMINALS & TANKS

4.1 General

Safe operation of facilities is a management responsibility. Management programs shall be established to assure conformance with applicable safety and operational standards, compliance with applicable regulations, and the use of appropriate work practices and procedures.

This section covers the elements most often used in the safe operations of terminals and tanks and provides guidance to assist this effort. This section addresses hazard identification, operating procedures, safe work practices, emergency response and control, management of change, training, prestart up safety and operational inspections, incident investigation, and contractor safety.

4.2 Hazard Identification

Programs shall be in effect and information on safety hazards shall be available to facilitate communication with employees, contractors, subcontractors, on-site vendors, emergency response personnel, and visitors. The information provided should be consistent with federal, state, and local hazard communication (29 CFR 1910) and Right-to-Know requirements and include provisions for ready access to Material Safety Data Sheets (MSDS).

Hazard identification information shall include operational hazards involved with the layout (see 2.1), specific equipment hazards, and other potential hazards, such as the release of a flammable liquid from a facility storage tank that has the potential for personal injury, environmental impact, or property damage.

In addition, hazard identification should consider the potential for reasonably foreseeable influences from outside the facility. These may include the following: natural occurrences (such as hurricanes, floods, and earthquakes), hazardous emissions from neighboring facilities, or physical hazards from traffic or neighboring occupancies.

The document 29 CFR 1910 contains the minimum federal requirements for communication of hazards to personnel. Other federal, state, and local regulations may apply.

4.3 Operating Procedures

Written operating procedures shall be developed to address the routine and nonroutine activities, such as emergency shutdown and startup that are conducted during the course of the facility's business. These procedures specifically include product storage and handling within the facility and all transfer operations. For those activities where significant deviation from procedures could lead to a loss of containment of flammable or combustible materials, the potential consequences should be highlighted and safeguards explained in detail in the procedure.

Management of change review is required before substantial revisions to operating procedures or the initiation of "temporary" operations (refer to 4.6).

4.4 Safe Work Practices

Written facility safe work practices (for example, hot-work, lockout/tagout, and confined space entry) applicable to employees, contractors, subcontractors, and on-site vendors shall be established. Safe work practices shall address safe conduct of administrative, operating, and maintenance activities as well as construction, removal, or idling activities (see 12.2). For new and modified facilities, these practices must be in place before startup or other work is performed.

Specific issues that shall be addressed by written procedures and permits are as follows: work authorization, hot work, confined space entry, lockout/tagout, and contractor orientation to owner safety rules and practices. Primary consideration in development of these procedures must be given to the health and safety of employees, contractors, and the public; the protection of the environment; and to the applicable regulatory requirements. Minimum requirements for lockout/tagout are identified in 29 CFR 1910 in a section entitled "The Control of Hazardous Energy (Lockout/Tagout)." Safe work practices for tank entry and cleaning are addressed in API Standard 2015. The minimum requirements for confined space entry are identified in 29 CFR 1910 in a section entitled "Permit-Required Confined Spaces." Refer also to API Publication 2009 for safe welding practices.

4.5 Emergency Response and Control Procedures

A written emergency response plan and procedures in compliance with relevant federal, state, and local regulatory requirements shall be in place for each facility to address response to anticipated emergencies. First aid medical treatment must be available, and response for major medical situations should be planned in advance.

Written plans for emergencies involving accidental release of hazardous substances shall address specific actions to be taken by employees and others, such as contractors, to control or mitigate the release or evacuate as appropriate. These emergencies include spills, fires, and abnormal conditions involving flammable and combustible materials. Refer to 5.6 for Fire Emergency Plan elements. Emergency or contingency plans are required by several federal regulatory agencies including 40 CFR 112 (SPCC) and OPA 90 Plans.

Emergency response procedures should address facility management and employee and contractor responsibilities including communication requirements both for internal purposes and for notification of and coordination with regulatory, governmental, public or mutual aid organizations.

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Emergency Response Procedures shall comply with 29 CFR 1910.

4.6 Management of Change

Systems shall be established for each facility to review potential health, safety, and environmental considerations resulting from proposed additions, modifications, or other changes that may periodically occur. Changes of the following nature should be properly managed by identifying and reviewing them before implementation:

a. Materials or products handled.
b. Equipment used or installed.
c. Operations and procedures.

The system should ensure that designs and operating procedures are reviewed before implementation and should revise as appropriate with the intent of minimizing adverse effects on the safety of the community, environment, and workforce. Minimum requirements for specific facilities are identified in 49 CFR 195 and 29 CFR 1910.

4.7 Training

All employer and contractor personnel working at the facility shall receive training appropriate to their duties and as required by the applicable regulations.

The primary responsibility for training lies with the employer of the employee. Training shall include pertinent portions of the operating, emergency response, and safe work procedures for that facility. Systems shall be established by each employer to ensure that their employees possess the required knowledge and skills to do their duties. Requirements for training and documentation of training for contract personnel shall be determined by agreement between the facility management and the contract employer.

Systems shall be established by the operators and contractors to assure conformance to the appropriate training requirements. Trainers shall be qualified, and training programs documented as required in the appropriate EPA, USCG, OSHA and DOT regulations. Periodic refresher training or revalidation of skills shall be incorporated in the overall training program as necessary or as required by regulations. API Recommended Practice 2220 contains additional guidance for contractor safety programs. At a minimum, training and documentation shall meet the applicable requirements of 29 CFR 1910 and other regulations to which the facility is subject.

4.8 Pre-Startup Safety and Operational Inspection

Pre-startup safety and operational inspection shall be conducted for new or modified facilities and equipment that involve a change that would likely have an influence or effect on the facilities' design or operation (the appropriate items from 4.1 to 4.7 should be reviewed). The pre-startup safety and operational inspection shall consider the following items, as appropriate:

a. Safety procedures and equipment are in place and function properly.
b. Construction or modifications meet design and manufacturer specifications.
c. Construction is in conformance with plans, codes, and applicable standards.
d. Operating and emergency response procedures are in place or have been reviewed and updated.
e. Any required training is complete.
f. Mechanical integrity and maintenance programs are in place or have been reviewed and updated, if necessary.
g. Equipment check out has been completed in accordance with manufacturer and company requirements.

If applicable, a management of change review (see 4.6) shall be performed.

4.9 Incident Investigation

Incidents and significant "near misses" should be investigated and the findings made available for use in prevention programs. The investigation should be conducted by personnel familiar with the facility work activities and specific work operations. Some incidents may warrant the formation of an incident investigation team that includes personnel skilled in investigation techniques and specialized technical skills.

The reporting, investigation, and documentation shall satisfy regulatory and, if applicable, insurance requirements. This process may include mandatory reporting requirements and requirements for specific personnel or job categories to be included as part of an investigation team. The goal of an investigation should be to determine "what happened" (the nature of the incident) and "why" (contributing causes) and to recommend the corrective actions or changes in procedures that the facility management may consider to prevent a recurrence. Records of the incident, investigation findings, recommendations, and actions taken (or explanations for inaction) should be kept on file at the facility. Consideration should be given to the benefits of sharing pertinent investigation findings and associated corrective action(s) with other similar facilities.

4.10 Contractor Safety

To help ensure the safety of the facility and all personnel, the use of contractors should be in accordance with procedures outlined in API Recommended Practice 2220. Additionally, contractor personnel shall adhere to the facility operating procedures, safe work practices (see 4.4), and training requirements (see 4.7). Minimum federal requirements for contractor safety are identified in 29 CFR 1926 and 29 CFR 1910.

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SECTION 5—FIRE PREVENTION AND PROTECTION

5.1 General

Fire prevention and protection is an important consideration for facilities handling flammable or combustible liquids. In a "hierarchy of importance," protection of personnel and the public, the environment, and equipment should dictate the amount of time and resources dedicated to identify, control, and mitigate potential incidents.

The practices and procedures addressed in this section are considered to be an effective means of preventing, controlling, and extinguishing fires. This section is specifically directed to fire prevention and protection provisions for tanks (Section 6) and loading activities (Section 9). Some of these provisions apply to the design and construction of new facilities or installations while others apply to operation, inspection, and maintenance for both new and existing facilities. Fire protection is not addressed here for other facility operations and structures, such as marine docks, warehouses, offices, and laboratories. Reference the appropriate NFPA code or other applicable standards, such as the ICBO *Uniform Fire Code*, for these structures, equipment, or operations.

5.2 Fire Prevention

5.2.1 VAPOR CONTROL

There is the potential for fire or explosion whenever a flammable vapor-air mixture exists. When handling flammable liquids with intermediate vapor pressure 1.5 to 4.5 pounds per square inch atmospheric (psia) [(10.3 to 31 kilopascal (kPa)] Reid Vapor Pressure and higher (for example, gasoline) and within 20°F (-6.7°C) of the flash point, flammable vapor-air mixtures may occur. Areas typically associated with flammable vapor-air mixtures include the following:

a. Within the vapor space of tanks.
b. Within barges, tank cars/trucks when loading or unloading.
c. Within portions of vapor recovery and vapor disposal systems.
d. Near the discharge of atmospheric vents.
e. Near a leak or spill.

Control of a flammable vapor-air mixture may be accomplished through these means:

a. Proper design and maintenance of facility system components.
b. Selection of storage tank type.
c. Use of safe work practices.
d. Although typically uncommon for large tanks, use of inerting or gas padding. Refer to NFPA 30, NFPA 69 and API Recommended Practice 2003 for further information on this subject.

5.2.2 CONTROL OF IGNITION SOURCES

Where flammable vapor-air mixtures may not be eliminated, sources of ignition must be controlled. Common ignition sources include the following: lightning, static electricity, stray currents, hot work, internal combustion engines, smoking, and improperly classified or protected electrical equipment.

Control of vapor ignition from lightning and static electricity is accomplished through proper design, maintenance, and operation as covered in API Recommended Practice 2003. Refer also to API Recommended Practice 500 for classifications of locations for electrical installations at petroleum facilities. Ignition from hot work is controlled by following established hot work permit procedures. Refer to 29 CFR 1910 for regulatory requirements for hot work and to 4.4 for discussion of safe work practices. Preventing ignition from electrical equipment is accomplished by these procedures:

a. Following appropriate electrical standards.
b. Assuring that equipment is in good operating condition, is properly installed, and is suitable for the area electrical hazards classification.

5.2.3 TANK OVERFILL PROTECTION

Accidental overfill of tanks that allows liquid to spill from the tank into the dike or surrounding area, particularly with flammable liquids, may create a potential for fires.

API Recommended Practice 2350 and NFPA 30 recommend specific overfill protection for terminals receiving flammable liquids from pipelines or marine vessels. Refer to 4.3 for the requirements for operating procedures, 6.1.9 for tank alarm requirements, Section 7 for dike berm requirements, and Section 9 for product transfer and control system requirements. Additionally, NFPA 30 addresses these and other issues related to the handling of flammable and combustible liquids.

5.2.4 INSPECTION AND MAINTENANCE PROGRAMS

Maintaining the integrity of storage tanks and piping systems containing flammable or combustible liquids is essential to the prevention of fires in and around tanks and terminals. Priority shall be given to correcting situations that would allow the release of vapor or liquid from the tank or piping system. See 6.2, 8.6, 8.7, 9.15, 10.2.2, 10.4.5, API Standard 653, and API Standard 570 for further information on inspection, maintenance, and testing.

5.2.5. HOUSEKEEPING

The dike area and the area around storage tanks shall be kept clean of combustible materials that potentially could be a source of fire exposure to the tank (see 7.2.11).

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5.3 Fire Fighting Equipment

Although fires in facilities are rare, the use of fire extinguishing equipment may be required by local regulatory agencies or by the facility owner. The emphasis should, however, be on fire prevention.

5.3.1 PORTABLE FIRE EXTINGUISHER

All facilities shall have portable fire extinguishers available at or near areas where fires may occur. Extinguishers shall be of suitable class and of an appropriate size for the nature of the fire that might occur. Refer to NFPA 10 for portable extinguisher classifications and requirements. Regulatory requirements for portable extinguishers are contained in 29 CFR 1910.

5.3.2 MOBILE AND PORTABLE FIRE FIGHTING EQUIPMENT

Larger facilities with trained fire brigades will often have specialized pieces of mobile and portable fire fighting equipment. Such equipment might include foam towers, large flow pumps and monitors, equipment trailers and foam trailers, and specialized fire fighting trucks. The nature and amount of such equipment will depend on local circumstances and the emergency plan. The need to provide fire fighting brigades and equipment should be evaluated for facilities where local services are inadequate.

5.4 Fire Extinguishment and Control

5.4.1 CONTROLLED BURN

With fires involving flammable and combustible liquids, rapid extinguishment may not always be possible or prudent. Where heat impingement on other vessels or facilities may be controlled and there is no danger to the public, it may be acceptable and sometimes safer to allow the fire to burn itself out under controlled conditions. This practice usually involves controlling the flow or otherwise limiting the amount of materials involved in the fire (pumping out the tank, for example) while providing cooling to surrounding equipment or structures that may be exposed to the heat of the fire or flame impingement. The decision to select this method of fire control should be made with the advice of qualified fire control personnel and the authority having jurisdiction (referred to as the incident command system).

5.4.2 MANUAL CONTROL & EXTINGUISHMENT

Manual fighting of flammable and combustible liquid fires usually involves the use of portable fire extinguishers on fires in their incipient or initial stages. Larger fires usually require the application of water for control and cooling or application of foam for fire extinguishment. For large fires, particularly those involving tanks, extinguishment requires special techniques, material, equipment, and trained personnel.

Manual attack of large fires should be supervised and conducted only by properly trained and qualified personnel, such as facilities that have trained, equipped, and qualified fire brigades or municipal fire departments. Refer to API Publication 2021 for additional information. The minimum requirements for industrial fire brigades are identified in 29 CFR 1910 and NFPA 600.

5.4.3 EXTINGUISHING SYSTEMS FOR TANKS

When used, fire extinguishing systems for tanks typically utilize fire fighting foams as the extinguishing agent. The need for these systems shall be consistent with the requirements of NFPA 30. Such systems require (a) an adequate water supply, (b) an adequate supply of correct foam solution or means of generating foam solution by mixing foam concentrate and water with a proportioner, and (c) a means of foam application to the tank in the proper amount and proportion.

The specific design of each system will vary based upon the size and type of tank being protected, the type of system (fixed or semi-fixed) on the tank, and the product being stored in the tank. The recommendations of NFPA 11 should be followed in the design and installation of foam systems for tank protection. Facility staff should be trained in the operation, maintenance, and testing of systems (refer to 4.7).

5.4.4 TRUCK/RAIL LOADING SYSTEMS

For tank truck or rail loading facilities, the need for fixed fire control or extinguishing systems should be based on a consideration of the specific risks involved. Systems, when provided, are usually water spray or foam-water spray systems. The systems and extinguishing agents are designed for the types of products loaded at the rack. The system nozzles are fixed and aimed to cover a predetermined area or surface with a calculated amount of water or foam (refer to 9.3 and 9.4). Facility staff, as well as drivers, should be trained in the system activation. Facility staff should be further trained in the operation, maintenance, and testing of the system (refer to 4.7).

These systems may be designed to activate automatically (through detection systems), manually, or both (automatic with manual activation). The detection system may include thermal or flame fire detection devices as well as combustible vapor (gas) detection devices.

NFPA 11, NFPA 15, and NFPA 16 should be consulted for the design and installation of these systems. Proper drainage of the area is necessary to prevent spreading of a fire by floating burning product towards other areas of the facility.

5.5 Fire Protection Water Supplies

The handling and storage of flammable or combustible liquids does not, in itself, require the presence of a fire protection water supply. The need is a function of the potential

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exposure to the public, employees, and the environment. Additionally, the risk to the facility, local jurisdictions, and the specific requirements to supply fire extinguishing systems or equipment may require a fire protection water supply.

The fire protection water supply should be based on consideration of the specific risks involved. Many facilities will need some emergency source of water, but specific needs will vary as some facilities are located in areas where inadequate or no water is available.

Fire protection water may be from any source that is capable of providing the required flow rate, at the necessary pressure, and for sufficient duration to either extinguish the largest expected fire or allow it to safely burn out by providing for cooling of exposed equipment and tanks. Such sources include utility water systems, public water supplies, dedicated in-plant fire protection water supplies, such as fire water ponds and water tanks or nearby natural water sources, such as rivers, lakes, and ponds.

The actual flow rate and volume of water required for a fire will depend upon the method of fire control and extinguishment desired as well as the type, amount, and size of fire systems, extinguishing materials, and equipment provided. The flow rate and pressure required should be capable of accommodating the equipment and systems expected to be operated simultaneously considering layout, spacing, and drainage for the single largest credible fire.

Additional information on related subjects may be found in the following standards:

a. NFPA 11 - *Low Expansion Foam and Combined Agent Systems.*
b. NFPA 15 - *Water Spray Fixed Systems.*
c. NFPA 20 - *Installation of Centrifugal Fire Pumps.*
d. NFPA 22 - *Water Tanks for Private Fire Protection.*
e. NFPA 24 - *Installation of Private Fire Service Mains and Their Appurtenances.*
f. NFPA 30 - *Flammable and Combustible Liquids Code.*

5.5.1 HOSE AND MONITORS

For those facilities that have an adequate supply of water available for manual fire fighting purposes, hose lines and monitors may be provided to supplement portable fire extinguishers. The water supply may be from any piped water system (refer to 5.5). The supply and pumping system shall be able to provide sufficient flow and pressure for the anticipated emergency. Hose lines and monitors should be used only by trained and qualified personnel.

5.6 Fire Emergency Plan

Each facility shall develop a written emergency plan that specifically addresses actions to be taken in the event of a fire in the facility. This plan may be developed as a separate plan or may be incorporated as a portion of other written emergency or fire plans for the facility (see 4.5). The written emergency plan shall address where appropriate each of the following elements:

a. Actions and responsibilities of employees in reporting a fire (see 4.7).
b. Responsibilities and action to be taken to control vapors (see 5.2.1) and prevent ignition (see 5.2.2) of vapors resulting from spills.
c. Actions and procedures (see 4.3) to be taken in fighting fires both manually and by activation and deactivation of fixed systems.
d. Desired method of extinguishment, such as through a controlled burn-out (see 5.4.1) or by application of an extinguishing agent.
e. Notification of appropriate authorities.

Fire emergency plans should be reviewed regularly and updated as products, equipment, and operating conditions change. Updating and filing of emergency plans shall be in accordance with the applicable federal, state, or local requirements.

Where the plan includes assistance from an outside organization, such as a public fire department or mutual aid group, the plan shall be previously coordinated with those organizations. Once the plan has been developed, employees must be trained on the duties and actions they are expected to perform. A training session and practice drill shall be conducted periodically or as per local/state regulatory requirements. Refer to NFPA 600 and 29 CFR 1910 Subpart L for employee and fire brigade requirements.

If outside agencies, mutual aid groups, or other organizations will be expected to respond to and assist in an emergency situation (see 4.5), drills should include their participation. Written records of training sessions (see 4.7) and drills shall be maintained as directed by the applicable regulatory requirements. Records should include the name and title of the employees and outside assistance personnel attending, the date of the drill or training session, the purpose of the drill, a description of the skills performed, and equipment failures or deficiencies.

5.7 Exposure Protection

Exposure protection for terminal and tank facilities is generally accomplished in the original design and construction by providing (a) adequate spacing between equipment (refer to Sections 2 & 8) and structures (refer to 11.1), (b) adequate drainage or impoundment (refer to Section 7) of spills or releases, and (c) separation from adjacent properties (refer to 2.3). Refer to NFPA 30 or other local recognized codes, such as the ICBO *Uniform Fire Code* for spacing requirements and for the requirements for dikes, berms, and drainage (refer also to Section 7). Burying and mounding of storage tanks is not recommended.

Fireproofing has limited application in these types of facilities. However, fireproofing should be considered for pro-

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tection of exposed structures, such as steel supports, whose failure could contribute to a fire.

Additionally, critical wiring, emergency shut down devices, isolation valves, or de-inventory capabilities (the ability to pump down a tank under fire conditions) should be protected from fire exposure.

5.8 Special Product Considerations

5.8.1 OXYGENATED PRODUCTS

Polar solvents and other water soluble liquids are a particular fire protection concern because they may be destructive to many of the standard fire fighting foams and may make extinguishment of fires involving these liquids more difficult.

With the recent requirements for oxygenated and reformulated fuels, larger quantities of polar solvents are being routinely handled in tank storage facilities and terminals. Fire fighting and extinguishment equipment should address the storage of oxygenated products, where applicable.

The most common polar solvent liquids found in tank and terminal facilities include alcohols, ethers, glycols, and amines. These oxygenates include methanol, ethanol, ethyl-tertiary-butyl-ether (ETBE), methyl-tertiary-butyl-ether (MTBE), and tertiary-amyl-methyl-ether (TAME). They may be handled in pure form or as a blend with gasoline. Generally, blends that have 10 percent to 15 percent polar solvent in gasoline may be treated as gasoline for fire fighting purposes. However, depending on the particular polar solvent, higher proportioning ratios (percentage of concentrate in water) and higher applications rates may be necessary.

Fire fighting and extinguishment of polar solvent liquids may require the use of special foams. These foams, sometimes referred to as "Alcohol Resistant," have been specifically formulated for use on polar solvents.

5.8.2 ASPHALT, CRUDE OIL, AND RESIDUAL FUEL

The light hydrocarbons may migrate out of crude oils and residual fuels and form a flammable mixture in the vapor space within the storage tank. Unless known to be otherwise, crude oil and residual fuels in storage should be considered to be a flammable liquid for fire fighting purposes.

Asphalt, crude oils, and residual fuels may have the characteristics that promote "boil-over" and "frothing." Unless known to be otherwise, asphalt, crude oil, and residual fuels in storage should be considered to have this potential. This is particularly true of those oils that contain a mixture of hydrocarbons with a wide range of boiling points.

5.8.3 GASOLINE AND DISTILLATES

Gasoline and distillates are considered to be stable and water insoluble. Normal fire fighting foams are suitable for extinguishing fires involving these liquids.

5.8.4 REACTIVE OR UNSTABLE CHEMICALS

Some chemical materials require special fire protection considerations due to the nature of their reactivity. This would include those liquids that react or decompose when heated or that would react spontaneously with air or water. In general, materials that react or decompose when heated will require a high application rate of water for cooling if exposed to a fire. If such products are stored on site, the adequacy of the fire fighting equipment (see 5.3), emergency action plan (see 4.5 and 5.6), and employee training (see 4.7) should be evaluated to determine the suitability of each to extinguish a fire involving these products.

The Material Safety Data Sheet (MSDS) for the material will provide specific information on its reactivity as well as recommendations on specific fire fighting and extinguishing techniques.

SECTION 6—TANKS

6.1 Aboveground Petroleum Storage Tanks

6.1.1 GENERAL

Design, construction, maintenance, inspection, and repair of aboveground storage tanks are covered in API Standard 620, API Standard 650, API Recommended Practice 651, API Recommended Practice 652, API Standard 653, and UL-142. The applicable mandatory provisions stipulated in these standards shall be followed. Where applicable codes or regulations are more stringent, such codes shall apply.

6.1.2 TANK SELECTION CRITERIA

Typical types of aboveground storage tanks found in petroleum terminals include fixed roof tanks, fixed roof tanks with internal floating roofs, fixed roof tanks with vapor recovery systems, external floating roofs, and external floating roof tanks with weather covers (aluminum domes).

Selection of tank types and required emission controls for storing volatile organic liquids (including petroleum liquids) in new or modified tanks shall conform to the EPA's *New Source Performance Standards*, 40 CFR 60 where required and meet the objectives of 3.3.1.

Factors that influence the type of aboveground tank ap-

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purtenances used for storage of a specific liquid include the following:

a. Vapor pressure, operating pressure, flash point, and temperature of stored product.
b. Tank capacity, frequency of turnover, and consequent value of vapor losses from filling and breathing.
c. Vapor control requirements, regulations, and new source issues. (API Publication 2517, API Publication 2518, and API Publication 2519 describe the evaporation losses from different types of aboveground storage tank designs).
d. Fire hazards, both to the tank and adjacent property (see Sections 2, 5, & 7).
e. Corrosiveness of stored product or its vapor to carbon steel (susceptibility of stored product to degradation, decomposition, or contamination due to the surrounding atmosphere that would lead to a corrosion problem) (see Section 10).
f. Static charge hazards (see 5.2.2).
g. Gas blanketing requirements.
h. Temperature and liquid level standards (see 9.12).
i. Product quality issues (minimize contamination by selection of design premises).
j. Provision for future flexibility for product changeovers that may have different specific gravities or other physical properties.
k. If used, tank seal material compatibility with the stored product.

6.1.3 RELEASE PREVENTION, LEAK DETECTION, AND AIR EMISSIONS

6.1.3.1 Release Prevention

The protection of the subgrade and ground water under new aboveground tanks is achieved through the use of a release prevention barrier (or other release prevention systems) (refer to API Standard 650, Appendix I). The use of inspection and maintenance practices described in API Standard 653 serve to supplement the protection of the subgrade and groundwater. High level alarms provide additional overfill protection (see 6.1.9).

Operating practices that ensure that inventories are properly monitored and controlled should be used to provide an additional method of detection of product leaks. There are a variety of additional practices that may be applicable for subgrade and groundwater protection and detection of product release. The effectiveness and applicability of each of the following practices depend upon the tank site conditions, tank history, stored fluids, and potential for a release to migrate beyond the facility boundaries:

a. Use of monitoring wells in the tank farm area.
b. Cathodic protection of the tank bottoms (see 10.4).
c. Use of planned and documented inspections (see 6.2).
d. Use of a leak detection system (see 6.1.3.2).
e. Use of a release prevention barrier (see 1.4.11) (for example, plastic geomembranes and clays) under the tank with site appropriate cathodic protection and leak detection.
f. Installation of a double steel bottom system that may include leak detection (see 6.1.3.2), corrosion protection/cathodic protection (see Section 10), and various structural support systems.
g. Smaller tanks set on a concrete slab.
h. Use of a reinforced liner per API specification (compatible with the products stored) installed over a sound steel tank bottom (per API Standard 653).
i. Use of a plan to manage the emptying of tanks and their repair (see 4.5).

6.1.3.2 Leak Detection

Many different types of leak detection systems currently exist. Some systems are designed to take continuous readings, and other systems use periodic readings (including visual monitoring) to determine tank integrity. Current API references on leak detection include API Publication 306, API Publication 307, and API Standard 650. Additionally, there are a number of different technologies being used, as well as new technologies being developed, for use in aboveground storage tanks to check integrity. Examples of systems include acoustic emissions, tracer methods, volumetric (including mass deviation) monitoring, and vapor sensing. Each system requires trained and skilled personnel to interpret results.

6.1.3.3 Aboveground Storage Tank Vapor Emissions

Emissions from aboveground atmospheric tanks storing petroleum and petroleum products are generated during product movements and normal tank breathing losses (collectively known as "working losses"). The following methods may be used to reduce emissions:

a. Installation of an internal floating roof inside a fixed roof tank in petroleum product service if the product vapor pressure is greater than 1.5 pounds per square inch atmospheric (psia) [10.3 kilopascal (kPa)] true vapor pressure (TVP).
b. Installation of a secondary seal (double seal) system on an external floating-roof tank.
c. Installation of a vapor control system for a fixed-roof tank in petroleum product service if the product vapor pressure is greater than 1.5 psia (10.3 kpa) TVP.
d. Vapor balance system.
e. Coating storage tanks white (or a light reflective color) for gasoline and distillates to reduce temperature changes thus minimizing losses and emissions. Refer to state and local agencies that may have differing requirements. Refer to 10.2 for coating requirements.
f. Adding a self-supporting fixed roof (dome) to an external roof tank (Refer to 3.4.1.1).

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In addition, if a facility meets the Clean Air Act definition of a "Major Source," then other emission requirements will apply. Refer to EPA's report, *Gasoline Marketing Industry (Stage I)—Background Information for Proposed Standards*, for details on EPA's emission requirements (see 3.3.1).

6.1.4 TANK CALIBRATION

Refer to API Standard 2550 and Chapter 2 of the *Manual of Petroleum Measurement Standards* for calibration of new or recalibration of existing tanks.

6.1.5 COATING AND PROTECTION FROM CORROSION

Refer to Section 10 of this standard.

6.1.6 TANK APPURTENANCES

6.1.6.1 Vents

Normal tank venting should be provided in conformance with API Standard 650 and API Standard 2000. Emergency venting equipment shall be provided for tanks that do not have a weak or frangible roof to shell connection (typically tanks less than 50 feet in diameter). Emergency vents may be required by local code or ordinance. A regular program for inspection and maintenance of vents should be conducted.

6.1.6.2 Gauge Hatches and Gauges

Tank gauging is a fundamental operation in accurately monitoring and controlling the integrity of aboveground storage tank inventories. The tank gauging system should be managed as a basic plant loss control device to analyze the inventory variations of each product. There are a variety of systems available ranging from the basic manual gauge to automatic remote reading systems. Available gauging systems are discussed in 6.1.6.2.1 through 6.1.6.2.3.

6.1.6.2.1 Conventional Tank Gauging System

Each tank should be equipped with a roof mounted manual gauge hatch. The hatch should be designed to be vapor tight against at least one ounce (0.28 Newton) internal force except asphalts and other services that are permitted to be open vented. The manual gauge readings are taken using tape and plumb bob to measure the liquid level in the tank. The gauge tube used in conjunction with the gauging system may be designed as an open or closed pipe. The open pipe or slotted gauge tube provides access for obtaining a representative sample in any level of the tank. The slotted tube is also a potential conduit for the transmission of vapors. Internal floats or other sealing devices may be required to contain these vapors. The nonslotted tube does not act as a conduit for vapors; however, it may not provide a representative sample of all tank levels.

Safe gauging practices should be established to prevent employee exposure and injury (refer to API Publication 2026).

6.1.6.2.2 Ground-Reading Gauges

Each new tank should be equipped with a gauge capable of being read at ground level. Gauge accuracy should be verified by periodic manual gauging.

6.1.6.2.3 Remote Gauging

Remote gauging systems are available using various types of technologies. The remote gauging system accuracy should be verified using manual gauges on a periodic basis.

6.1.6.3 Overflow Slots

Cone roof tanks with internal floating roofs may be equipped with overflow slots to prevent damage to the fixed roof and internal floating roof in the event of an overfill condition.

6.1.6.4 Manholes

Manholes shall be provided in the tank shell and in the cone roof of covered floating roof tanks to provide access for inspection and repair. Entry into such areas shall be in accordance with the facility's Confined Space Entry procedure (refer to 4.4 and 29 CFR 1910).

6.1.6.5 Ladders

Internal ladders may be provided in covered floating roof tanks extending from the manhole in the fixed cone roof to the floor. The ladder shall be designed for full travel of the floating roof regardless of any settling of the roof supports. The use of interior tank ladders on external floating roof tanks (with or without a geodesic dome) is an option that should be reviewed from a safety standpoint. Some tanks are not equipped with a permanent ladder to ensure that the tank entry process is controlled. Safety procedures must be established and followed when using these systems (see 4.4). Refer to API Standard 650 Appendix H.

6.1.6.6 Anti-rotation Devices

Anti-rotation devices are required in open and covered floaters to prevent rotation of floating roofs.

6.1.6.7 Bonding

Requirements for bonding an internal or external floating roof to the tank shell are described in API Standard 650 and Recommended Practice 2003.

6.1.6.8 Grounding

Aboveground storage tanks shall be capable of dissipating

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electrical charges to ground (refer to NFPA 30, NFPA 77, and NFPA 780).

6.1.6.9 Thermometers

Accurate measurement of product temperature is another factor in maintaining accurate inventories. There are a number of ways to install thermowells and thermometers in a tank. Examples include thermometers installed in the tank shell, on an internal floating suction, or submerged in the tank gauge tube. Portable thermometers may also be used to measure product temperatures at various levels. A representative temperature is required to accurately determine the net quantity of stored product. Safe work practices should be established for use of thermometers to prevent release of a static electric spark or employee exposure.

6.1.6.10 Stairs and Platforms

Stairs, ladders, and platforms are provided for safe personnel access to the top of storage tanks. The design requirements for these appurtenances are covered in API Standard 650 and 29 CFR 1910. These items are typically supplied by the tank fabricator. Design options include the use of different types of non-skid surfaces, stairways or ladders, and platforms for gauging, temperature measurement, and sampling. Railings and toeboards are also part of the design requirements. See 29 CFR 1910, for stairway, ladder, and toeboard requirements.

6.1.7 WATER DRAW-OFF

A water draw-off system to remove and properly dispose of water (and off specification product) that accumulates inside the bottom of the tank shall be provided. The typical system includes a drain line near the bottom of the tank shell that is equipped with a freeze resistant (where weather conditions warrant) steel valve. A tank water bottom collection system may be provided that uses hard piped connections, portable tank trucks, or other alternatives depending upon site logistics, product characteristics, permit requirements, and exposures (for example, freezing temperatures). Tank water draw off systems are not normally used for hot oils (for example, asphalt and greases). Safety procedures and safe work practices should be established to minimize exposures and to control the draw off of product during water draw off operations. Provisions should be in place to ensure that water draw off valves are kept closed and secured when not in use to prevent product release.

6.1.8 ROOF DRAINS

External floating roof tanks shall be equipped with roof drains. The drains are used to remove the accumulation of water from the roof. The drains are typically located at the center of the external floating roof and routed through the contained liquid in the tank by hose or flexible pipe connection. Discharge from external floating roof drain systems should be carefully managed to prevent product releases.

6.1.9 TANK ALARMS

6.1.9.1 General

High level-alarms provide additional overfill protection for receiving tanks. Local regulations may require the installation of high-level alarm systems (see 5.2.3). API Recommended Practice 2350 details the use and types of high-level alarm systems. NFPA 30 defines the requirements for tank overfill protection including the requirements for high level alarms. Liquid level alarm systems may also be used to provide low level protection for pumps and other equipment. Procedures and practices for regular testing of alarm systems, including the primary sensor, should be established.

6.1.9.2 System Requirements

The following features represent the typical requirements for a two stage high level alarm system for an unstaffed terminal:

a. One of the two alarms should be independent of any tank-gauging device. See 6.1.9.3 for required alarm stages.
b. The alarm should be tied to an annunciator and located where personnel on duty throughout the transfer of product may promptly arrange for flow stoppage or diversion.
c. The alarm system shall be electrically supervised.

6.1.9.3 Alarm Stages

Tank level alarms should have two stages. The first stage, high, is set to give a warning at a specified number of minutes before liquid reaches the high level target for the tank. The second stage, high/high, alarm should be set to give a warning at a specified number of minutes before maximum fill height occurs. The alarms, both audible and visual, should be designed to notify appropriate facility personnel or product transfer (such as pipeline) personnel.

The safe distance between the alarms and the high fill level target varies in each tank because of different tank sizes, seal arrangements, maximum product flow rates into the tank, and the time necessary for personnel to perform an orderly shutdown. The safe distances between alarm stages and high fill level should be established at each facility on a tank-by-tank basis taking into account the above conditions (refer to API Recommended Practice 2350).

6.2 Operations, Inspections, Maintenance, and Repair for Aboveground Tanks

6.2.1 GENERAL

Safe and environmentally sound management of aboveground tanks include inspections made in accordance with

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API Standard 653, development and use of formal operating procedures (see 4.3), and training of operators (see 4.7) to execute these responsibilities. The following list may be used as the basis for an operations plan:

a. Physical inventory procedures including reconciliation.
b. Inspections and checks of the following:
   1. Rectifier readings for cathodic protection (see 10.4.4).
   2. Calibration of thermometers and automatic tank gauges.
   3. Fire protection equipment and systems (see 5.3).
   4. Overall tank farm operations and conditions.
   5. External and internal condition of the tanks per API Standard 653 and regulatory/permit requirements.
   6. High-level alarm system.
   7. Dikes, drainage valves, berms, and walkways.
   8. Roof drains on external floating roof tanks and water drawoffs on all tanks.
   9. Vapor holder tank above the bladder for vapors in tanks so equipped.
   10. Tank vents and flame arresters.
   11. Internal and external floating roofs and seals including the straps bonding the roof to the shell.
   12. Periodic settlement monitoring (refer to API Standard 653).
   13. Differential pressures on any strainers or filters to establish need for cleaning.
   14. Operation of the floating suction in each tank.
   15. Railings, stairs, platforms, and gauging stations and positions for compliance with OSHA and related safety considerations.
   16. External corrosion (see 10.2 and 10.4).
   17. Dike area for drainage adequacy, erosion, and weed control (see 7.2.3 and 7.2.11).
   18. Inspections required by the SPCC regulations (40 CFR 112).
   19. Tank mixers.
c. Review of the spill contingency plan for the tank farm and update/revalidation as required by regulation (see 4.5).
d. Monitoring the observation wells and record data.
e. Identifying tanks and other equipment for maintenance planning, operations, and records.

The operations plan should describe acceptable conditions and actions appropriate for the site specific facilities. In addition to the points listed above, the plan should have other safety considerations including the following: hot work permits, tank watch procedures, and other preventive measures (see Sections 4 and 5). Records should be kept of these activities.

6.2.2 REQUIREMENTS

All aboveground tanks shall be inspected, maintained, repaired, and tested in accordance with the mandatory provisions of API Standard 653. Tank inspectors shall be certified in accordance with API Standard 653. Tank entry and cleaning shall be conducted in accordance with API Standard 2015.

6.3 Fiberglass Aboveground Storage Tanks

Refer to API Specification 12P for minimum requirements for aboveground fiberglass storage tanks. The use of fiberglass tanks would normally be limited, due to many considerations including lack of fire resistance. Class 1, 2, or 3A liquids may not be stored in tanks made of combustible materials. Refer to NFPA 30 for more information. Products, such as water, water treatment, fire fighting foam, wastes, bulk lubricants, and specific types of chemicals including nonflammable corrosives are typically stored in fiberglass tanks.

6.4 Underground Tanks and Piping

The following API documents cover the recommended practices and design considerations for underground tanks and should be consulted for detailed requirements:

a. API Recommended Practice 1604.
b. API Recommended Practice 1615.
c. API Publication 1621.
d. API Recommended Practice 1631.
e. API Recommended Practice 1632.

The use of underground tanks for petroleum product storage should be considered in light of the federal underground storage tank regulatory requirements as described in the EPA technical standards 40 CFR 280 as well as state and local requirements. However, underground tanks for nonproduct storage will continue to be used for loading rack spill containment and drainage systems.

6.5 Underground Storage Tank Vapor Emissions

Emissions from underground tanks storing gasoline are generated during the filling of the tank. Stage I vapor controls are widely used to reduce emissions generated during tank filling. In addition, to prevent normal breathing losses some state and local jurisdictions require the installation of a pressure-vacuum (P-V) valve on the vapor-vent line.

For further information concerning Stage I vapor controls refer to API Recommended Practice 1615 and to EPA's report, *Gasoline Marketing Industry (Stage I): Background Information for Proposed Standards.*

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SECTION 7—DIKES AND BERMS

7.1 Overview

This section covers the design and construction of dikes and berms for tank containment areas. Operating features associated with dikes and berms are addressed in 7.2.10, 7.2.11, 7.3.4, and 7.3.5.

7.2 Dikes

7.2.1 GENERAL

When used for secondary containment of tank contents, dikes and intermediate dikes shall be constructed in accordance with NFPA 30. Consideration shall also be given to the requirements in 29 CFR 1910 and 40 CFR 112 and applicable state and local regulations or permits. These documents contain information on siting, spacing, design, and other aspects of tanks and secondary containment and should be consulted for special considerations in the fabrication of containment dikes.

7.2.2 CAPACITY

Dikes shall be sized so as to contain the greatest amount of liquid that may be released from the largest tank within the diked area assuming a full tank and providing sufficient allowance for precipitation after taking into account protection provided by remote impoundments (see NFPA 30).

7.2.3 DRAINAGE OF RAINWATER

Unless other provisions are made for drainage, the floor of the diked area shall be graded to at least 1 percent for 50 feet (15.24 meters) away from the tank(s) or to the dike base, whichever is less.

7.2.3.1 The sloped area shall be directed toward one or more drain openings or retention areas. Major paths of drainage should be routed so that piping, equipment, tanks, or vessels will not be seriously exposed should flammable or combustible liquid in the drainage ditch ignite. Piping should be protected where it crosses over drainage routes or ditches.

7.2.3.2 Unless other provisions are made in the site's SPCC plan, drainage that would bypass the in-plant treatment system shall be accomplished through block valves that are located or that may be safely operated during an emergency from outside the diked area. These valves shall be normally closed and secured. Drainage system designs should consider the fire water loading as well.

7.2.3.3 Dike drainage shall be documented where required by the SPCC regulation (40 CFR 112).

7.2.4 HEIGHT

Dike height shall be limited to an average of not more than 6 feet (1.9 meters) above the interior grade unless special provisions are made for normal access and necessary emergency access to tanks, valves, and other equipment with safe egress from the diked area provided. Provisions made for vehicular and foot traffic shall not compromise the integrity of the secondary containment.

7.2.5 WALKWAYS

On earthen dikes 3 feet (1 meter) or more in height, a flat section at the top not less than 2 feet (0.6 meters) wide shall be provided.

7.2.6 SLOPE

The slope of the side walls of earthen dikes shall be consistent with the angle of repose of the dike material with consideration given to safe maintenance operations.

7.2.7 LINERS

Consideration shall be given to the permeability of the dike material. In those areas where the material lacks sufficient impermeability, the use of liners should be explored and evaluated from an engineering basis. Liners may be a synthetic material, a composite of natural substances, or other material capable of meeting the impermeability requirements for the site conditions.

7.2.8 OPERATING EQUIPMENT

Valves used for fire fighting shall be located outside of the diked area to provide protection from and access to during a fire or a spill. While location of pumps outside the diked area is desirable from a fire safety standpoint, location of pumps inside a diked area is attractive for protection of the environment. Both safety and environmental factors should be weighed when planning the location of pumps, of operating valves, and of valves used for fire fighting. Refer to 8.3 through 8.7 for pipe and pump requirements.

7.2.9 PIPING

Piping or conduit passing through dike walls shall be designed to prevent buildup of excessive stresses on pipe or conduit as a result of settlement. Penetration of dikes (except drains) should be kept to a minimum to eliminate leakage paths through the dike. The area surrounding the penetration shall be sealed with a high temperature fire resistant material to prevent migration of liquids through the dike. Piping that penetrates the dike should be protected from corrosion by use of coatings, pipe wraps, or secondary containment (refer to 8.3, 8.4, 10.2, and 10.4). Conduit passing through the dike should be sealed as it leaves the dike wall.

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7.2.10 RESTORATION

If the integrity of a dike used for secondary containment around an in-service tank is compromised, such as by cutting the dike to gain access with heavy equipment, the dike shall be restored to a height or width necessary to maintain the integrity of the containment system before the area is left unattended unless other provisions are made to provide for the necessary containment (refer to 4.4 and 4.10).

7.2.11 WEED CONTROL

Measures shall be taken to ensure that any vegetation adjacent to or within the diked area is kept to a level that does not pose a threat to successful fire control (see 5.2.2, 5.2.4, and 5.2.5).

7.3 Berms

7.3.1 GENERAL

For the purpose of this standard, a *berm* is defined as the annular area around the tank, inside the dike, normally used for access to the tank and the equipment surrounding it.

7.3.2 CONSTRUCTION

Berms shall consist of a walking area, sloped gently away from the tank, of sufficient width and extending sufficiently around the circumference of the tank so as to permit access to hatches, equipment, and appurtenances on or near the tank, such as mixers and valves.

NFPA 30 calls for a minimum distance of 5 feet (1.52 m) between the tank and the toe of the dike for certain cases, and the standard should be consulted where this may be a factor.

7.3.3 GRADING

Grading between the berm and the toe of the dike shall allow drainage as covered in 7.2.3.

7.3.4 RESTORATION

If the berm is disturbed for any reason, it should be restored to original condition as soon as practical.

7.3.5 EROSION CONTROL

Where necessary, erosion control measures should be considered for application around the tank perimeter.

SECTION 8—PIPE, VALVES, PUMPS & PIPING SYSTEMS

8.1 General

Pressure piping and related components at facilities covered under this section shall conform to the provisions of ASME B31.4 or ASME B31.3, as appropriate, or other codes referenced in this standard. Marine systems may have additional codes to conform to, such as AWS D-1.1 Welding in Building Construction or ABS Rules for Building and Classing Steel Vessels for load conditions.

The guidelines in this section are applicable for liquid hydrocarbons that have low corrosivity and that are at temperatures between -20°F (-29°C) and +190°F (88°C). Special design and operating considerations must be made for liquid hydrocarbons that are highly corrosive or that operate outside of this temperature range. Water, air, and other utility piping systems that are not designed for hydrocarbon usage are covered in 11.2.

8.2 Material Compatibility

With the advent of new products, including oxygenated fuels, special consideration shall be given to the compatibility of the piping system, especially seal materials, gaskets, and other elastomers with additives and oxygenates, such as TAME, ETBE, MTBE, ethanol, and methanol.

8.3 Piping Systems

8.3.1 GENERAL

A piping system consists of the pipe, fittings, valves, pumps, and other types of equipment that are connected together to provide a means of conveyance to handle product. While each component of the system may have its own specifications and design parameters, care must be taken to consider the interaction of the components in the overall piping system design.

The design of the piping system should be as uncomplicated as possible minimizing connections, fittings, and valves. Operating errors, maintenance costs, and risks of an environmental release will increase as a piping system becomes more complex. The designer shall consider provisions for the integrity assurance of the piping system (refer to 8.6).

8.3.2 ABOVEGROUND VERSUS BURIED PIPING SYSTEMS

There are advantages and disadvantages to both aboveground and buried piping systems. Factors to be considered in placement of piping include the following:

a. Local regulations may favor either aboveground or buried piping and should always be consulted.
b. Aboveground piping is accessible for visual inspection, maintenance, and repair, thus enhancing leak detection and prevention.
c. Access to areas of the facility during emergencies may be inhibited with aboveground piping.
d. Piping modifications are usually easier to complete on aboveground piping due to the absence of excavation requirements.

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e. Aboveground piping will be more affected by radiant heat. Therefore, consideration should be made for expansion and pressure-relief on closed sections.
f. Aboveground piping provides more flexibility for movement in all planes. Flexibility may be necessary to accommodate uneven settlement, shifting foundations, soil movement, earthquakes, movements from line shocks or water hammer, and movements from thermal expansion and contraction.
g. Buried piping may be necessary to completely drain tanks, such as storage tanks, sumps, drop-out tanks, and other vessels.
h. Buried steel piping may require protection from external corrosion (for example, effective coating and cathodic protection).
i. Buried piping is less susceptible to damage by vandalism, vehicular traffic, and fire. However, buried piping should be marked or mapped to avoid damage during excavation work.

8.3.3 ANCHORING STRUCTURES AND PIPE SUPPORTS

Anchor structures and pipe supports shall be designed and installed to support or control pipe movement where appropriate, and thus protect equipment, such as pumps, tanks, and valves from excess mechanical loading. Anchor structures and pipe supports shall be constructed to prevent wear and corrosion of the piping and should be designed to allow adjustment of the support. Refer to ASME B31.3 and ASME B31.4 for technical aspects of anchoring and supporting piping. Fire protection provisions should be considered, and are covered in NFPA 30 and other locally recognized codes, such as the ICBO *Uniform Fire Code*.

8.3.4 PIPE INSULATION & HEAT TRACING

High pour point hydrocarbons may require heat tracing or insulation. Design considerations should include these factors: temperature requirements, thermal expansion, clearance for insulation at pipe supports, insulation type, and insulation protection (weather and mechanical). The designer should review the potential for corrosion (see 10.2.3, 10.2.4 and 10.4.3 for design considerations) and the risks of corrosion under the insulation as noted in API Standard 570.

8.3.5 DEAD LEGS

Industry experience has shown that dead legs (sections of pipe that do not experience flow under normal conditions) may be subject to internal corrosion and freezing or rupture problems. For these reasons the construction of dead legs should be avoided.

8.3.6 VAPOR CONTROL PIPING SYSTEMS

Pipe for vapor control systems may include spiral weld pipe that conforms to API Specification 5L as well as pipe listed in 8.4. The design of a vapor control piping system should include providing proper slope towards condensation collection points. Refer to API Publication 2557 (for storage and transfer operations) for technical considerations. For marine installations (refer to 33 CFR 155).

8.3.7 DRAINAGE SYSTEMS

Design of drainage systems should include consideration of material compatibility and pressure limitations for the worst case situation as well as the possibility of uncontrolled flow of flammable hydrocarbons through a drainage system. See 11.2.3.2 and 11.2.3.3 for additional comments.

8.3.8 PACKAGED PIPING SYSTEMS

Piping that interconnects individual pieces or stages of equipment within a packaged equipment assembly shall be in accordance with ASME B31.4 or ASME B31.3, as appropriate. In addition, packaged refrigeration equipment may conform to the requirements of ASME B31.5.

8.3.9 FLOW DIAGRAMS

Easy-to-follow flow diagrams which indicate all major valves and flow directions for normal conditions as well as for upset conditions should be available at each facility. Valves may be numbered or otherwise identified to match the flow diagram.

8.4 Piping Components

8.4.1 PIPE

The recommended types of pipe include seamless, electric-resistance-welded (ERW), and electric-fusion-welded pipe (EFW) as listed in ASME B31.4 and ASME B31.3. Manufacturing specifications are also included in the specifications listed in ASME B31.4 and ASME B31.3.

Calculations for the diameter of the pipe are normally based on desired liquid flow velocity and pressure drop in the pipe. Typical recommended design rates are between 5 and 20 feet per second (1.52 to 6.10 meters/second). Higher velocity flows may result in static electric buildup, hydraulic surges, or internal pipe erosion, requiring additional design evaluation. Refer to API Recommended Practice 2003 for additional information.

Steel pipe that is subject to external corrosion shall be protected (see 10.2 and 10.4). The wall thickness of the pipe should be selected based upon pipe diameter, pipe grade, allowable stress, operating pressure and temperature, corrosion effects, dynamic effects (vibration, hydraulic shock), weight effects (pipe, pipe contents, snow, insulation), and movement of connected components. Formulas for wall thickness calculations are included in ASME B31.4 and ASME B31.3, as applicable.

Typical piping joints are welded, flanged, or threaded. While piping joints are necessary, piping systems should be

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designed to minimize the number of joints, and joints should be welded wherever practicable. As a general guideline, joints on piping 2 inches (5.1 cm) and larger should be welded, or flanged joints on piping 2 inches (5.1 cm) and smaller may be threaded. Requirements for nondestructive testing of welded joints are listed in ASME B31.4 and ASME B31.3, as applicable.

8.4.2 FITTINGS AND FLANGES

Fittings shall be constructed of forged steel and shall have a minimum pressure rating of 2,000 pounds per square inch (psi) (13,790 kPa) for threaded fittings and 3,000 psi (20,684 kPa) for socket-weld fittings. Cast iron and brass fittings shall not be used in new construction or as replacement components for liquid hydrocarbon service. Aluminum fittings are commonly used on truck loading racks for dry-break couplers and breakaway fittings, but if used, they should be used downstream of shut-off valves. In addition, the possibility of structural failure of aluminum fittings during a fire should be considered when such fittings are used at the truck loading racks.

Miter welds should be avoided except for certain applications. Refer to ASME B31.4 and ASME B31.3, as appropriate for miter weld use information.

Threaded fittings and socket-weld fittings shall conform to ASME B16.11, and flanged fittings shall conform to ASME B16.5 and ASME B16.47. Welded fittings shall conform with ASME B16.9, ASME B16.28, and MSS SP-75. Threaded fittings should not be buried. The use of buried flanges should be minimized.

The use of solid steel plugs is recommended over hollow core plugs, and the plugs should conform to ASME B16.11.

Pipe joints dependent upon the friction characteristics and integrity of combustible materials for mechanical integrity or liquid tightness of piping shall be evaluated on an engineering basis before being used for transporting petroleum liquids.

Union fittings with machine finished metal to metal seats or with spiral wound gaskets are recommended. Insulating unions with cathodic protection gaskets may be used as appropriate. Union fittings shall conform to MSS SP-83 and ASME B16.11.

8.4.3 GASKETS & BOLTS

Gasket material should be compatible with the contained fluid. Gaskets shall meet the fire resistant properties as specified in NFPA 30 and shall conform with ASME B16.20 or ASME B16.21.

Studs used with flange joints shall extend completely through the nuts and shall conform to ASME B16.5 for studs and ASTM A193, ASTM A194, or ASTM A325 for nuts.

8.4.4 VALVES

A piping system designer has a wide assortment of valves to choose from. The type of valve to be used will vary for each specific situation. Lists of applicable standards for the various valves are included in ASME B31.4 and ASME B31.3 as appropriate. Information labels on valves should be protected and not removed, not painted over, or otherwise damaged. Consideration during valve selection should be given to minimizing hydrocarbon vapors emitted from the valve. See 3.3.1.4 regarding fugitive emissions.

Fire-tested valves should be considered for use in certain locations, such as the first valve from the tank and the isolation valves in the loading areas. Fire-test specifications include API Specification 6FA and API Standard 607. Valve size may be a consideration in determining the need for a fire-tested valve.

Nonductile cast iron valves as well as low melting temperature metals, such as brass and aluminum (see 8.4.2) shall not be used in new construction or as a replacement for valves in hydrocarbon service. Bolted bonnet valves are preferred over union or screwed bonnet valves.

Where product separation or bubble-tight closure is critical for a particular situation, double block and bleed valves should be used.

Valves used in isolation or shut-off service should be able to maintain a positive seal of liquid when the piping system is disconnected on the discharge end of the valve. Wafer valves or valves with bolts extending from flange to flange outside of the valve body should not be used for isolation service without a special means of fire protection. This precaution is due to the possibility of bolt deformation during a fire. See 5.7 for additional information on exposure protection.

Check valves are used to control the direction of flow and may not be relied upon for positive shutoff in the reverse direction. Use of check valves should be considered at the discharge of each pump on a multi-pump piping system and on the pump bypass piping at centrifugal pumps.

Pressure relief valves should be considered to prevent excessive pressure buildup in a closed piping system, such as around closed valves, check valves, or pumps. If pressure relief valves are used, a regular documented testing program shall be instituted for each pressure relief valve.

Discharge from a pressure relief valve should be directed to a lower pressure receptacle where the relieved liquid will not interfere with downstream conditions. Block valves on a pressure relief system should be secured open to ensure a properly functioning system.

Remote operation of valves may be utilized for convenience as well as for emergency operation. Consideration should be given to the final valve position in a failure mode (for example, fail-open, fail-close, or fail-unchanged).

8.5 Pumps

8.5.1 GENERAL INFORMATION

Centrifugal and positive displacement (PD) pumps are typically used in facility applications. Centrifugal pumps

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typically conform to ANSI/ASME B73.1M or API Standard 610. PD pumps typically conform to API Standard 674, API Standard 675, and API Standard 676. Pumps conforming to specific trade group standards may be considered for use. For nonasphalt hydrocarbon service, the pressure-containing components of a pump installed during new construction or for replacement shall not be constructed of cast iron.

Pumps should be selected on performance specifications, compatibility with the service application, durability, and anticipated maintenance requirements. The pump and motor selected should be reviewed to ensure that motor horsepower is adequate for all performance points over the entire range of operation, not just one design point. Pump and motor nameplates and instrumentation should be protected and should not be removed, painted over, or otherwise damaged.

8.5.2 PUMP INSTALLATION

Piping systems at pump locations should be designed to allow for pump removal and maintenance. At locations with multiple pumps, each pump should have isolation valves and check valves. When check valves are used, the check valve should be designed so as not to slam during alternating pump cycling (see 8.4.4).

Positive displacement pumps should utilize a pressure relief system and may require pulsation dampeners. When designing pressure relief systems, the design should account for the additive nature of pressure relief in the system.

The piping system should be designed to prevent binding, misalignment, and seal wear on the pump as well as to avoid structural limitations of the pump casing. Proper piping design techniques, anchoring of pipes and pumps, and stress analysis are preferred over flexible connections to the pump.

If a pump is installed within a contained area, the motor shall be of a class compatible with that environment (see 11.2.2). Drip and spill containment around the pump should be considered.

8.5.3 PUMP OPERATION & MAINTENANCE

The operator should maintain a data file for each pump that should include pump make and model, motor make and model, flow rate, pressure rating, pump curves, dimensional data (pump, motor, shaft, stages) seal, and wearing material data. The design information should be readily accessible and should be updated to include any changes in service, use, or pump specifications. Changes in service should initiate a review of the pump data to ensure the existing design is suitable for the new service.

8.6 Pipeline Integrity Assurance of Existing Piping Systems

8.6.1 GENERAL

All facilities shall periodically assure the integrity of system piping. The integrity of buried piping systems shall be assured by a comprehensive program utilizing a combination of the methods in 8.6.3. The integrity of aboveground piping systems shall be assessed by one of the methods in 8.6.2. Further details are provided in ASME B31.4 and API Standard 570. Requirements for marine facilities are included in 33 CFR 156.

8.6.2 ABOVEGROUND PIPING SYSTEMS

Methods of integrity assurance for aboveground piping include visual inspection, vapor detection, ultrasonic wall thickness measurement, and pressure testing.

8.6.3 BURIED PIPING SYSTEMS

Methods of integrity assurance for buried piping (both gravity and pressure) include the following: pressure testing, volumetric testing, acoustic testing, selective excavation and inspection, visual inspection of pipe whenever exposed, ground penetrating radar, tracer testing, monitoring of the cathodic protection system (see 10.4), instrumented internal inspection device, visual inspection of the area surrounding the buried piping, and monitoring wells.

8.6.4 PRESSURE TESTING MEDIUM

Pressure testing may be performed with the contained fluids under these conditions: at pressures up to the maximum normal operating pressure of a piping system or up to the design pressure if original test data is available or up to a proposed test pressure that will produce a hoop stress of less than 35 percent of the specified minimum yield strength of a piping system for which original test data is not available. For pressure testing at pressures above the maximum normal operating pressure but below the original design pressure when original test data is not available and the test pressure will produce a hoop stress in excess of 35 percent of the specified minimum yield strength of the piping system, the test shall be performed using water or hydrocarbons with a flash point above 120°F (49°C) unless other special precautions are taken. The temperature of the pipe and the test medium shall be in accordance with the recommendations of ASME B31.3 and B31.4, as appropriate. Water should not be used to pressure test hot oil or asphalt piping. See 33 CFR 156 for facilities under Coast Guard jurisdiction.

8.7 Testing Following Construction

Testing of piping systems following new construction or major modification to piping system components shall conform to API Recommended Practice 1110, ASME B31.4, or ASME B31.3, as appropriate. See 33 CFR 156 for facilities under Coast Guard jurisdiction. For piping conforming to ASME B31.3, refer to API Standard 570 for piping system testing.

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SECTION 9—LOADING, UNLOADING AND PRODUCT TRANSFER FACILITIES

9.1 Scope

This section covers facilities that transfer all classifications of liquid petroleum to or from tank trucks (excluding oil production facilities), rail tank cars, marine vessels, and stationary storage tanks. Refer to 1.4.3, 1.4.5, and 5.8 of this standard for classifications of petroleum liquids.

9.2 General Design

All transfer facilities shall be designed in conformance with applicable local and national codes and good engineering practices. In the design process, particular attention should be given to the specific hazards associated with tank trucks, rail tank cars, and marine vessels entering transfer facilities. Design and maintenance practices should consider impact loads and potential damage from docking operations at marine facilities. For additional guidance, refer to the ICOS/OCIMF/IAPH *International Safety Guide for Oil Tankers and Terminals*. Consideration should be given to the need for protective barriers or other means of hazard identification (see 4.2) to minimize potential for vehicular accident damage to facilities.

Loading, unloading, and product transfer areas should be located in relationship to other facilities to provide room for safe ingress/egress. An ingress and egress plan should be developed that will provide safe movement of truck or rail tank car traffic while providing for adequate staging areas. Care should be taken to not block normal traffic flow. Safe driving speeds should be established, posted, and enforced. Refer to 2.3 for additional information on spacing of facilities.

Due to the nature of the materials being handled, transfer facilities shall incorporate adequate fire prevention design criteria and environmental protection systems. Refer to Section 5 for more guidance on fire protection systems, to Sections 3 and 7 for information on environmental issues, and to Section 6 for tanks.

Only noncombustible materials shall be used for the construction of loading racks. Piping, valves, and fittings that are to be used shall be designed and fabricated from materials compatible with the product being handled and having adequate strength and durability to withstand the pressures, stress, and exposures to which they may be subjected. Refer to Section 8 for additional guidance.

Loading arms and components should be designed to maintain sound mechanical and structural integrity. Heat traced and jacketed loading arms and hoses should incorporate handling accessories to prevent personnel from contacting high temperature surfaces. Hoses used for loading should be supported in such a manner as to prevent personnel injury and prevent damages or excessive wear.

Special consideration shall be given to the design and location of loading arms and hoses to assure adequate safety clearance is maintained when in the stored position. Rail clearance guidance should be obtained from the carrier servicing the facility or from the Association of American Railroads (AAR).

Pumps and loading devices shall be sized to provide rates of flow appropriate to the design capacity of the facility and its safe operation. Refer to API Recommended Practice 2003 and NFPA 77 for guidelines on filling rates. Care shall be taken during design to ensure that an operator, when present, may follow the course of loading and unloading at all times and have adequate time to shutdown the transfer. Control systems, including automatic set stop valves, shall have adequate time to shutdown the transfer before tanks are filled beyond maximum safe fill levels (see 6.1.9). Where flow control valves or emergency shutdown valves are employed that use line pressure to operate, the system hydraulics shall be designed so that the required pressures are available under all operating conditions to allow for a safe and reliable shutdown (see 4.5). In general, fail safe control valves are recommended for use in the control system. Where automated systems are employed that require electrical power, the systems shall be designed to safely shutdown operations in the event of a power failure. Design considerations should include development of an area lighting plan (see 11.2.2 and 11.2.5). Refer also to the *IES Lighting Handbook*. Lighting levels should be provided consistent with the safe operational needs of each specific area of the facility to facilitate loading, unloading, and product transfer.

9.3 Truck Loading/Unloading

Further reference for truck loading may be found in 2.3.3, 3.3.1.1, and 11.3.3.

9.3.1 SPILL CONTAINMENT PAVING

Adequate spill containment shall be incorporated into the drainage design to contain and control a spill resulting from accidental overfill, equipment failure, or malfunction.

Truck loading rack areas should include concrete pavement with a raised edge (curbing) or other spill containment method provided around the loading rack perimeter. The raised edge should be sloped or rounded to facilitate truck access. Concrete joints should be sealed with petroleum resistant sealant to prevent leaks to subgrade. Pavement should be sloped toward catch basins and drains that are piped to containment or treatment facilities.

Catch basins should be located so that the liquid will flow away from the tank truck, loading rack equipment, and personnel. Catch basins should be accessible under fire conditions and away from the actual loading operation. Drainage systems should be designed to prevent the spread of fire

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from one loading lane to another and to areas outside of the loading rack. Manholes with fire seals should be incorporated to safely flush and drain away any spills.

Maintenance of drainage systems shall include necessary measures to prevent snow, ice, dirt, or other solids from clogging the system. For further reference see 11.2.3 and 11.3.

9.3.2 LOADING RACK CANOPY

Where loading operations are conducted in areas that are exposed to the elements, measures to minimize the introduction of surface water and precipitation into the loading rack containment system will reduce the quantity of liquid that must be environmentally treated if stormwater segregation has been provided (see 3.3.2). Installation of canopies over the entire loading area is an effective means of reducing the volume of precipitation entering the loading area containment system.

Canopies should be constructed of noncombustible materials in accordance with all local, state, and federal standards (see 11.1). Special care must be taken to assure safe personnel egress in case of emergencies.

Proper canopy lighting should be provided to allow safe operation and to reduce the likelihood of accidents causing spills or product mixing. Lighting under the canopy should provide adequate illumination to read meters, conduct required operations, and perform general maintenance of the loading rack equipment. Refer to the *IES Lighting Handbook* or applicable state or local codes for lighting requirements. The electrical classification of the canopy lighting system shall adhere to the requirements stipulated in 11.2.2.

9.3.3 LOADING ARMS, HOSES, AND METERS

When possible, the loading arms and hoses should be arranged to enable the filling of all tank compartments without having to move the vehicle, thus reducing the potential for accidents.

Loading arms and components must have high mechanical integrity to prevent emissions of VOCs under normal operating conditions. The use of aluminum pipe, fittings, or components should be limited to quick connect couplings, breakaway couplers, and drop tubes. Loading arms and hoses should be inspected and maintained, as necessary, to ensure integrity of the system (see 8.4).

Pumps capable of pressures in excess of the safe working pressure of the hose or loading arm shall be provided with bypasses, relief valves, or other arrangements to protect against excessive pressure. Pressure relief systems shall be employed when necessary to prevent or control over-pressurization resulting from thermal expansion or line shock. Relief devices shall be tested periodically to verify their set pressure (see 8.4.4). Records of this testing should be maintained at the facility.

9.3.3.1 Top Loading

Top loading is most applicable for distillate products, asphalt, or liquid sulfur. However, top loading is discouraged, particularly for flammable liquids, due to the potential to generate a higher rate of vapors as well as the potential hazard of falling.

Measures shall be taken to protect personnel from falls during top loading operations. Where platforms are provided, stairways and handrails serving the loading rack shall be maintained in good condition and shall comply with applicable OSHA regulations. Refer to 4.4 for additional safety requirements.

Drop tubes of adequate length and proper design should be used to reduce splashing, static discharge, and vapor generation (see 3.3.1.1 and 5.2).

Loading procedures should be established that incorporate safe loading practices (see 4.3). Signs displaying the required loading procedures should be mounted so as to be visible from each loading position.

9.3.3.2 Bottom Loading

The use of the bottom loading method is strongly encouraged for light products. In addition to the reduction of air emissions, there is improved safety because all work is conducted at grade level, and splash generated static electricity buildup and discharge of a spark is reduced.

Refer to API Recommended Practice 1004 for recommendations concerning selection of standardized equipment, such as couplers, adapters, layout of connections, secondary shutoff controls systems, and vapor recovery systems.

Loading procedures should be established that incorporate safe loading practices. Signs displaying the required loading procedures should be mounted so as to be visible from each loading position.

A system to prevent exit from a loading rack while a loading arm or vapor hose is connected, such as an air brake interlock system, should be considered.

9.3.3.3 Heavy Products Applications

High viscosity/heavy products may impose additional system requirements for loading/ unloading. Purging, heating, or pigging systems may be necessary for specific products or facility requirements. Many times these products are sold by weight, and truck scales are used for measurement and loading control. Heavy product applications apply for both top and bottom loading systems. Individual product specifications should guide the loading system configuration.

9.3.4 CONTROL SYSTEMS

Control Systems incorporated in the design of truck loading facilities range from complex computer controlled systems to manual operation. As a minimum, the following controls should be incorporated into the design:

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9.3.4.1 Grounding/Bonding

Static electricity may be generated during the loading/unloading operation from splash loading or from designs that promote static generation, such as high flow rates, filters, and water entrainment. It is recommended that the truck be bonded to the loading rack by a continuous monitored bond connection before connecting loading hoses and initiation of the loading process. In addition, the absence of a bond connection to the truck should prevent loading by a control system interlock. Bonding allows for the equalization of electrical charge between the truck and the rack, thus reducing the chance of a spark when the loading arm is moved away from the truck.

The loading rack should be grounded to earth either through piping or ground rods. Additional information and requirements on grounding and bonding for product transfers is contained in API Recommended Practice 2003 and NFPA 30.

All electrical equipment and systems shall be grounded in accordance with the latest version of the *National Electrical Code* (NFPA 70) (see 11.2.2).

Metal structures, such as canopies and supports over loading/unloading areas shall be bonded to the rack and grounded to minimize damage that might occur from direct-strike lightning (see 11.1). Additional information on lightning protection may be found in API Recommended Practice 2003.

9.3.4.2 Overfill Protection

When top loading a tank truck, valves shall be of the self-closing type and manually held open except when automatic means are provided for shutting off flow when the vehicle or compartment is full. Automatic shutoff systems shall have a manual means of shutoff if the automatic system fails.

When bottom loading a tank truck, a positive means shall be provided for loading a predetermined quantity of product together with a secondary automatic shutoff control to prevent overfill. Primary protection should be provided by meter controlled automatic valves. The secondary automatic shutdown devices should include devices such as thermistor or optical level probes. These devices should be of self-checking design. The components of safety systems on the vehicle and the loading rack shall be compatible. Precautions that prevent overfilling, product contamination, and unsafe conditions shall be established and followed. See Section 4 for additional safe operation requirements. Refer also to Section 8 and API Recommended Practice 1004 for equipment standardization and additional recommendations.

9.3.4.3 Fire Prevention and Protection

Fire prevention measures, such as emergency shutdown, positive drainage, overfill protection, and reduction of the potential for static build up should be considered during the design of loading and unloading systems (see 5.2).

In addition to a manual means of activating a fixed fire-protection system, an automatic activation system should be considered and in some cases may be required by code. Refer to 5.4.4 for additional information concerning fire protection in loading/unloading areas.

9.3.4.4 Metering and Flow Control

Product metering for truck loading/unloading operations may be accomplished using various types of metering equipment. Most often either a positive displacement meter or a turbine meter is used. The selection should be based on product characteristics and facility operating design. For high viscosity products, mass flow meters or vehicle scales are generally used, and the product is loaded by mass in lieu of volume. Measurement equipment must be calibrated to maintain accuracy and in many cases is required by law.

Starting and stopping of the loading operation may be done manually; however, use of automatic set stop controls are preferred.

The use of automatic set-stop controls with low flow start capabilities should be considered when the possibility of a flammable atmosphere could occur. Flammable atmospheres may result from the vapor pressure of the material being handled or may be the result of switch loading operations (when a low vapor pressure product is loaded into a compartment that contains a flammable vapor from previous use at or above the lower flammable limit). Having a flammable atmosphere during loading introduces a possible ignition hazard unless the loading is begun at a low flow rate that will minimize the generation of a static charge. Additional precautions that should be exercised when loading with a flammable atmosphere are described in API Recommended Practice 2003.

9.3.5 ADDITIVE INJECTION FACILITIES

When additive injection is required, measures should be taken to ensure that the proper amount is injected. Calibration of additive injection systems should be conducted in accordance with equipment manufacturers' and the company's recommendations. Additional verification may be warranted either through metering or positive indication of the piston stroke (if the injection system utilizes a calibrated cylinder). The additive system shall be designed to be compatible with the chemical being injected.

Operating practices to ensure that inventories are properly monitored and controlled shall be used.

9.3.5.1 Additive Storage

Dikes/berms, secondary containment, gauging, and other methods of spill prevention and control shall be considered for additive storage. See Sections 3, 6, and 7 of this standard for guidance.

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9.3.5.2 Additive Transfer

Additive transfer is the same as described in 9.3.1 through 9.3.4. Spill containment within the immediate area of the hose coupling activities should be considered. Provisions for lighting, grounding, and other safety precautions should be considered.

9.4 Rail Tank Car Loading and Unloading

Rail tank cars are loaded and unloaded in a similar manner to truck loading. There are two types of loading: (a) loading an individual rail tank car, and (b) using a unit-train that has numerous tank cars interconnected that act as one large tank.

When rail tank cars are top loaded, measures shall be taken to protect personnel from falls since work is above grade level. Where platforms are provided, stairways and handrails serving the loading rack shall be maintained in good condition and shall comply with applicable OSHA regulations. When top loading, drop tubes shall be of sufficient length to reach the bottom of the tanks being loaded and designed to minimize splash.

When loading or unloading rail tank cars, vapors should be handled in accordance with Section 9.7.

Pressure relief facilities should be considered for the pumping system as well as the rail tank car.

9.4.1 SPILL CONTAINMENT

Spill containment for loading/unloading of rail tank cars should be made of a material capable of containing the spilled product. Spills should be diverted to a product collection system (see 11.2.3). Spill containment systems are similar to those used with truck loading racks (see 9.3.1).

9.4.2 RAIL TANK CAR LOADING AND UNLOADING CANOPY

The installation of a canopy at the rail tank car loading and unloading area depends upon facility specific operational requirements and weather conditions. If it is determined that a rail tank car loading/unloading canopy is required, refer to 9.3.2 and 11.1 for guidance.

9.4.3 LOADING ARMS, HOSES, AND COUPLERS

Loading arms, hoses, and couplers used for top loading rail tank cars are similar to those used in the tank truck loading. Bottom loading will generally require use of hoses and couplers to match specific rail tank car connections. Refer to 9.3.3 for general guidance and AAR *Mechanical Division Standards*.

9.4.4 CONTROL AND SAFETY SYSTEMS

Control Systems incorporated in rail tank car loading facilities are similar to those for truck loading; Section 9.3 of this standard should be reviewed before design.

9.4.4.1 Grounding and Bonding

It is recommended that a means of preventing the buildup of an electrostatic charge and dissipating any charge generated on the rail tank car be incorporated in the design of rail tank car loading facilities. See API Recommended Practice 2003 for guidance. Bonding should be provided between the rail tank car and loading rack to equalize the electrical potential. Rail tank cars are usually grounded by contact of the wheels to the rails.

9.4.4.2 Overfill Protection

Overfill protection should be included in the design of rail tank car loading systems. Systems similar to those discussed in 9.3 of this standard should be considered.

9.4.4.3 Fire Protection

If fire protection facilities are to be installed, refer to 5.4.4 for guidance.

9.4.4.4 Metering and Flow Control

Except for the unit-train configuration of rail tank car loading, the metering and flow control systems used are similar to those used in truck loading. Refer to 9.3 for further guidance.

9.5 Marine Loading/Unloading

Additional information on marine loading/unloading operation and safety may be found in the ICOS/OCIMF/IAPH *International Safety Guide for Oil Tankers and Terminals*.

9.5.1 SPILL PREVENTION AND CONTAINMENT

Loading of barges and marine vessels (tanker ships) shall not commence until the shore supervisor and the person in charge of the vessel agree that the barge or marine vessel is properly moored and all connections are secure. Communications shall be maintained between the barge/marine vessel and shore during the loading/unloading operation (see 33 CFR 26). See API Recommended Practice 1125 for guidelines for overfill control systems for barges and marine vessels.

Where barges or marine vessels are loaded or unloaded through articulated arms or hose connections, spill containment shall be incorporated to contain and control a spill resulting from leakage or accidental equipment malfunction depending upon the vapor pressure of the material being handled. The terminal should incorporate storage for containment booms and keep deployment equipment ready to contain a spill on the water.

9.5.2 LOADING ARMS AND HOSES

Marine swivel joints, when used, shall be designed so that

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if the packing materials should fail, the mechanical strength of the joint will not be impaired.

Marine hoses or loading arms shall be capable of accommodating the combined effects of change in draft and tidal change. Mooring lines shall be kept adjusted to prevent sway and surge movements of the vessel from placing stress on the cargo transfer system.

Shut-off valves shall be provided for each flexible product line at the base of the cargo transfer equipment or near the approach to the pier or dock to stop the flow in the event of rupture. This may include pressure actuated valves that close automatically in the event of a leak.

Loading arms and hoses shall be designed and tested periodically in accordance with OCIMF, *Design and Construction Specification for Marine Loading Areas* and United States Coast Guard (USCG) requirements (33 CFR 156). Test pressure is typically 125 percent of the maximum operating system pressure. Records of the tests shall be maintained.

9.5.3 CONTROL SYSTEMS

The control systems that are used in the design of marine loading and unloading facilities are similar to those used in truck loading and unloading systems. See 9.3 of this standard for reference.

Emergency shutdown procedures and notifications shall be in accordance with the applicable U.S. Coast Guard Regulations. Adequate lighting shall be provided for both low level area lighting and work level lighting in equipment and loading areas (see 11.2.2).

9.5.3.1 Grounding and bonding

Barges and marine vessels are inherently grounded as they sit in water. For additional information on grounding and bonding of tank ships and barges, refer to API Recommended Practice 2003. The use of insulated flanges on pipe connections for loading/unloading hoses should be considered to reduce the chance of an arc when hoses are disconnected.

9.5.3.2 Overfill Protection

For overfill protection of marine vessels and barges, refer to API Recommended Practice 1125.

9.5.3.3 Fire Prevention and Protection

For fire protection of marine vessel and barge facilities refer to NFPA 30.

9.5.3.4 Metering and Flow Control

Metering and flow control for tank ship and barge loading and unloading is similar to truck loading and unloading (see 9.3.4.4, API Recommended Practice 2003, and API Recommended Practice 1125).

9.5.4 ADDITIVE INJECTION FACILITIES

Additive injection during marine transfers will incorporate equipment for injection and storage. Many of the same procedures and practices applicable to truck loading rack additive injection are valid in marine transfers. See 9.3.5 for further information.

9.5.5 DREDGING

Periodic soundings should be conducted to verify adequate water depth for the vessels that use or could use the dock. Dredging shall be conducted as required. All dredging and disposal shall comply with the applicable requirements of the authority having jurisdiction, such as the Army Corps of Engineers (Section 404 dredging permitting), state port authority, or EPA.

9.6 Aviation Loading/Unloading

For aviation fuel activities, refer to API Publication 1500.

9.6.1 SPILL CONTAINMENT

Adequate spill containment shall be incorporated into the drainage design to contain and control a spill from accidental overfill, equipment failure, or malfunction. Refer to truck loading spill containment (see 9.3.1).

9.6.2 LOADING CANOPY

Refer to truck loading rack canopy (see 9.3.2).

9.6.3 LOADING ARMS, HOSES, AND METERS

Refer to truck loading arm, hoses, and meters (see 9.3.3 and API Publication 1529). Filters/Separators information can be found in API Publication 1581.

9.6.4 CONTROL AND SAFETY SYSTEMS

Refer to the Truck Loading Rack Control and Safety System (see 9.3.4).

9.6.5 FIRE PREVENTION AND PROTECTION

Refer to Section 5 of this standard and NFPA 30.

9.6.6 ADDITIVE INJECTION FACILITIES

If additives are required to be injected into the fuel, refer to 9.3.5 of this standard for requirements.

9.7 Vapor Control

Vapor control requirements are governed by EPA, state, and local air quality agencies. Requirements vary greatly from one region to another within the United States. Consultation with applicable agencies is advisable before design, construction,

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and operation of new vapor control systems, vapor combustion units, or modifications to existing systems (see 3.3.1).

Volatile organic compounds (VOC) and hazardous air pollutant (HAP) emissions may be generated when tanks, tank trucks, rail tank cars, and marine vessels are filled with petroleum and petroleum products. Existing vapors that may be in the containment space at the time of loading are displaced as well as any vapors created by the material being loaded. Technologies for recovering or incinerating these emissions during loading operations are reviewed in 9.7.1.4. Extensive guidance is available in API Recommended Practice 1124 and API Publication 2557.

9.7.1 TANK TRUCK VAPOR CONTROL

During tank truck loading, displaced gasoline vapors should be collected and absorbed, condensed or thermally oxidized, as appropriate (see 9.7.1.4). If the facility tankage is equipped with vapor control, returning vapors to tankage should be considered. If switch loading (see 9.3.4.4) has occurred, trucks loading fuel oil may also emit gasoline vapors that should be processed. In the event the vapor control unit shuts down or is not operating properly, procedures should be provided to shut down the loading facilities unless given authorization to continue by the regulatory authority having jurisdiction. Connections to the vapor recovery system shall be designed to prevent the escape of vapor to the atmosphere when not connected to a vehicle.

Tank truck unloading operations discussed in this standard apply to product deliveries to aboveground storage tanks. Tank truck unloading operations for deliveries to underground storage tanks is covered in API Recommended Practice 1615.

9.7.1.1 Submerged Filling

All loading operations should be performed using submerged filling. Submerged filling is the introduction of liquid into the tank truck below the liquid level. Submerged filling minimizes droplet entrainment, evaporation, turbulence, and static electricity generation.

Submerged filling may be accomplished when top loading, by lowering the fill tube until it is just off of the bottom of the tank and filling slowly until the fill tube is submerged. In bottom loading, the fill is started slowly until the product covers the area where a fixed fill pipe enters the tank truck from the bottom. Bottom loading is the preferred method. The automatic loading rates should be slow until the filling nozzles are submerged. When bottom loading, the coupler between the liquid loading hose and the truck shall be a dry break disconnect.

9.7.1.2 Rail Tank Car Vapor Control

Collection of vapors during rail tank car transfer is similar to truck tank transfers (see 9.7.1).

9.7.1.3 Marine Vapor Control

Emissions from loading/unloading operations are controlled at the vapor outlet of the vessel compartment or tank being filled.

The use of marine vapor control systems during loading or ballasting of bulk liquid cargo tanks introduces potentially significant hazards different from those at tank truck and rail tank car loading facilities. Regulations governing the safe design, installation, and operation of marine vapor control systems include 33 CFR 154 and 46 CFR 39.

The primary hazards associated with the use of marine vapor control equipment are cargo tank over- or under-pressurization, overfill and spillage, fire, explosion, or detonation.

See API Recommended Practice 1124 for guidance on vapor collection manifolds.

9.7.1.4 Vapor Processing System

The major types of vapor processors include the following: carbon adsorption, refrigeration, thermal oxidation, lean oil absorption, or a combination of such systems.

A vapor processing system consists of the following basic components:

a. A vapor processor (recovery unit or combuster).
b. Gasoline supply or return systems for the processor (not applicable to thermal oxidation).

Note: Gasoline return lines should be located on the tank as far from suction lines as possible to allow for dilution of Vapor Recovery Unit (VRU) return gasoline.

c. Vapor handling lines.
d. Vapor-tight tank trucks (rail tank cars or marine vessels/barges) constructed or modified for bottom loading and vapor recovery or top loading and vapor recovery.

It is recommended that the accumulation of vapors within the explosive range be minimized.

9.7.1.5 Vapor Handling Systems

Vapor handling systems may be regulated at some terminals and bulk plants under State Implementation Plans. Refer to appropriate federal, state, and local regulations. A tank truck rack vapor handling system includes the following components:

a. A closed system with no operating vents open to the atmosphere during tank truck loading or unloading.
b. Vapor-tight gasoline tank trucks properly equipped for bottom loading and vapor recovery (or for top loading and vapor recovery).
c. Vapor recovery arms or hoses at each gasoline truck loading station.
d. A vapor collection system from the loading rack and truck unloading station to the gasoline storage tanks. Check valves shall be installed in truck vapor connection branches at loading racks.

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e. Vapor balance connections at the gasoline storage tanks should typically be located at an existing roof nozzle or modified roof manhole cover.
f. A vapor tank may be used as a means to reduce unit cycling and reduce energy consumption of the vapor control unit(s).
g. Flame arresters or detonation devices should be considered for installation between loading facilities and vapor control unit(s). For marine vapor recovery refer to 9.7.1.3.

9.7.1.6 Inspection and Maintenance

All vapor recovery processing equipment shall be subject to periodic maintenance, testing, and inspections. Records of inspections and maintenance should be kept on file at the facility. Periodic testing of vapor return lines should be conducted to verify the tightness of connections.

When flame arresters or detonation arresters are installed in vapor lines, they should be periodically inspected and cleaned if needed to ensure that clogging that could cause elevated pressures and malfunction of the device in an emergency has not occurred.

Pressure/vacuum vents should be inspected periodically and serviced if necessary to ensure the absence of clogging and proper operation.

Other required testing of vapor recovery systems, including the vapor recovery unit or vapor oxidizer, should be conducted in accordance with the manufacturers' recommendations, facility procedures, and the applicable regulatory requirements.

9.8 Oxygenate Blending

Ethanol, methanol, MTBE, ETBE, and TAME are oxygenates commonly in use today. MSDSs shall be consulted for hazards involved in handling alcohol and other oxygenates. Refer to API Recommended Practice 1626 and API Recommended Practice 1627 for guidelines on the storage and handling of ethanol/methanol and other cosolvent products. Refer also to 5.8 for fire protection guidelines.

Alcohols and other oxygenates may be blended into gasoline at truck loading racks sequentially (one component loaded into the truck after the other) or by the inline or ratio methods (all component streams loaded in the proper ratio simultaneously into the truck). With the sequential method, one meter is typically used for all components. A separate meter is used for each component when inline blending.

When oxygenates are blended with gasoline, procedures should be in place to assure that the proper volume correction factors (VCF) for the components or the blend being measured are used.

When blending by the sequential method with one meter, procedures should be in place establishing meter factors for all components if product characteristics or differing flow profiles cause meter factors to vary.

Care should be taken in the design of oxygenate blending systems to ensure equipment compatibility with the oxygenate being handled. Valves, hoses, gaskets, meters, and pumps should be specified to be compatible with the oxygenate to minimize maintenance problems (see 8.2).

9.9 Emergency Shutdown Systems

Emergency shutdown systems should be provided at all product transfer facilities. Emergency shutdown systems should include clearly marked manual activation switches at loading/unloading areas and at remote areas that are accessible during an emergency. When fire protection or vapor detection systems are activated, they should automatically activate the emergency shutdown system.

The emergency shutdown system should shutdown all flow and provide a visual or audible indication to personnel in the area as well as supervisory personnel as appropriate. If the emergency shutdown system automatically deactivates the main circuit breaker for the facility incoming power, consideration should be given to providing emergency backup power for fire protection, gate operation, and lighting for safe evacuation of the area at night (see 11.2.2).

Consideration shall also be given to the orderly shutdown of all systems upon the activation of the emergency shutdown system (see 4.5 and 5.6).

Design of emergency shutdown systems for marine facilities shall be in accordance with U.S. Coast Guard Regulations 33 CFR 154.

9.10 Product Testing

Testing to verify product quality should be conducted in compliance with regulatory, operational, and contractual requirements and may be conducted at the beginning, during, or after a receipt and during or after any product blending operation. For marine transfers, quality testing should be completed before the start of transfer, conducted periodically during transfer, and conducted upon completion of transfer.

9.11 Communication

A reliable means of communication or notification shall be provided, such as voice intercoms, alarms, signals, telephones, or approved radios between the truck loading rack, rail tank car, loading facility, or marine vessel dock and with personnel who may respond in the event of an emergency when product transfers are in progress (see 11.2.5). In the case of marine terminals, Coast Guard Regulations (33 CFR 154) shall be followed.

9.12 Measurements

When the transfer involves custody transfer or if the quantity loaded must be known with a high degree of accuracy, measurement of the transfer should be accomplished according to all appropriate chapters of the API *Manual of Petroleum Measurements Standards* and applicable weights

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and measures requirements.

When the metering systems require the accuracy of custody transfer, consideration should be given to the method and to any required additional facilities or services to prove each custody transfer meter.

9.13 Valves, Lines, Loading Arms, and Hose Product Identification

When more than one product is handled within the facility, the transfer equipment should be adequately marked with product type or so designated that the operator may identify the various loading arms, hoses, lines, or valves without having to trace them back to their source or destination (refer to 10.2.11).

9.14 Release Prevention System in Loading/Unloading Areas

In all new construction or major modifications, consideration should be given to the installation of release containment and drainage systems where truck, marine, or rail transfers take place.

9.15 Maintenance/Testing

Periodic inspection and preventive maintenance should be conducted on all transfer systems to control leaks.

Accurate inventory records may be maintained and periodically reconciled for indication of possible leakage from tanks and piping systems.

Other maintenance to be considered includes the inspection and cleaning of strainers and filters, proper adjustment of flow control and shutdown devices, and the maintenance of any other systems required to safely operate the facility.

All marine, rail tank car, and truck loading facilities should be maintained in a neat and orderly manner, and no accumulation of trash or other combustible material shall be permitted except in approved containers.

Hotwork in areas where vapors may be present, such as welding or cutting or use of spark-producing tools shall be permitted only under adherence to safe work practices that are designed to prevent accidental ignition of petroleum vapors (see 4.4).

Complete maintenance records should be maintained by the operator for all equipment within a terminal.

9.16 Auxiliary Systems

For pump back and own use fuel systems that are located separate from truck loading racks, similar spill containment, electrical, lighting, and grounding/bonding requirements should be considered (see 9.3).

SECTION 10—CORROSION CONTROL

10.1 Scope

Section 10 covers the information, procedures, and practices for achieving effective corrosion control for terminal and tank facilities in hydrocarbon service. This section contains provisions for the application of cathodic protection, coatings, and linings to existing and new storage tanks and other terminal facilities and structures. Detailed designs and specifications are not provided. Such designs should be developed by persons knowledgeable in corrosion control practices and in accordance with applicable codes and regulations. This section does not designate specific practices for specific situations, as varied conditions preclude standardization of such practices.

This section presents corrosion control in three areas: protective coatings (10.2), interior tank lining (10.3), and cathodic protection (10.4). Other corrosion control requirements, such as the use of nonmetallic materials and the problems of bimetallic corrosion are not presented. A thorough discussion of corrosion mechanisms may be found in various publications including the following: API Recommended Practice 651, API Recommended Practice 652, NACE International Recommended Practice 0169-92, and ASM International's *Metals Handbook*, Volume 13, *Corrosion*.

10.2 Protective Coatings

The use of protective coatings should be evaluated during the design phase of work. Protective coatings are applied to structures for a variety of reasons including these:

a. Protection of the structure from corrosion.
b. Appearance.
c. Regulatory requirements.
d. Safety and operating efficiency.

Coatings provide protection to structures through any one or more of the following processes:

a. Prevent contact between structure and environment.
b. Limit contact between structure and environment.
c. Release inhibitors to mitigate corrosion.
d. Produce a protective or electrolytic current (for example, zinc rich coatings).
e. Provide passive nonconductive films to electrically isolate the structure.

10.2.1 NEW OR COMPLETE RE-COAT

New construction and total recoating of existing structures require that the structures be cleaned sufficiently (according to at least the manufacturer's recommendations or better) before the coating system is applied. The coating system

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should be specified for the particular product and environment to which the structure is exposed. The use of lead and other heavy metal based coatings should be avoided.

10.2.2 MAINTENANCE OF COATINGS

The facility preventive maintenance program should address the maintenance of coatings. Maintaining coatings will extend the life of the system at costs lower than total removal and recoating. Maintenance may include the following: topcoating, touch-up, and other means of extending the life and providing protection. Maintenance coating is effective only if the existing coating is adherent and in good condition. Coating must be compatible to ensure long life and good performance.

10.2.3 COATING SYSTEM EVALUATIONS

Evaluations of the coating systems on all structures should be conducted periodically. These evaluations should include the following items:

a. Coating thickness.
b. Condition and failure identification.
c. Existing system type vs. changes in environment or product.
d. Adhesion.
e. Severity of any rusting/corrosion.

Evaluations should be performed by a knowledgeable person with experience in coating systems. Evaluations should include a recommendation on what action, if any, needs to be taken to maintain or replace the coating system. The evaluation process should include a prioritization process to rank the structures according to the quality and needs of the coating system and the consequences of coating failure.

10.2.4 COATING SELECTION

The selection of coating systems may be guided by the following factors:

a. Environmental and product resistance.
b. Appearance.
c. Safety.
d. Surface preparation requirements.
e. Application and maintenance requirements.
f. Substrate to be coated.
g. Design life.
h. Accessibility for future repair.
i. Consequences of coating failure.
j. Lead and heavy metals content

10.2.5 APPLICATION PROCEDURES

10.2.5.1 Surface Preparation

Surface preparation should be accomplished using the equipment necessary to achieve the desired profile and standard. Surface preparation should be in accordance with the referenced SSPC specifications or NACE International Standards. Safe work or hot work permits should be issued as required (see 4.4).

10.2.5.1.1 Degreasing

Oil and grease should be removed from all surfaces to be coated or painted before subsequent surface preparation. Solvent cleaning should be done using a scrub and rinse process in accordance with SSPC Specification SP-1, *Solvent Cleaning*.

10.2.5.1.2 Manual Cleaning

Surfaces prepared by manual methods should be cleaned by removing all loose paint, mill scale, rust, dirt, oil, grease, moisture, and other detrimental foreign matter by scraping, chipping, sanding, needle guns, wire brushing, or power buffing in accordance with SSPC Specification SP-1 *Solvent Cleaning*, SP-2 *Hand Tool Cleaning*, and SP-3 *Power Tool Cleaning*.

10.2.5.1.3 Blasting Specifications

All tags, labels, tank name plates, glass covers, pilot lights, fire equipment, fusible links on fire safety valves, fire system detectors, swivels, meter heads, valves with open stems and stem housings, rubber seals, pressure-vacuum vents, printer drives, motors, pumps, or other delicate equipment, and other nonpaint surfaces, such as aluminum, brick, tile, insulation, and transite should be completely covered at all times while blasting is in progress to protect against both direct blast and media entry. Blast finishes include the following:

a. Brush Off Blast Cleaning—This specification is further expressed and qualified by the SSPC Specification SP-7, *Brush Off Blast Cleaning*. Visual appearance of the surface may be that indicated for NACE International No. 4 per Standard TM-01-70.
b. Commercial Blast Cleaning—This specification is further expressed and qualified by the SSPC Specification SP-6, *Commercial Blast Cleaning*. Visual appearance of the surface may be that indicated for NACE International No. 3 per Standard TM-01-70.
c. Near White Blast Cleaning—This specification is further expressed and qualified by the SSPC Specification SP-10, *Near-White Blast Cleaning*. Visual appearance of the surface may be that indicated for NACE International No. 2 per Standard TM-01-70.
d. White Blast Cleaning—This specification is further expressed and qualified by the SSPC Specification SP-5, *White Metal Blast Cleaning*. Visual appearance of the surface may be that indicated for NACE International No. 1 per Standard TM-01-70.

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When blasting is performed with blast grit containing silicia, appropriate protective measures must be taken. Grit must be properly collected and disposed.

10.2.5.2 Lead Paint Removal

Lead pigment has been a paint component for many years and has been widely used for coating piping and storage tanks. Eventually, lead based systems become embrittled, lose adhesion and may need to be removed. Removal of these coatings using conventional methods (hand tool, power tool, and blasting) creates friable dust that may be respirable and pose a health hazard to exposed persons. Consequently, lead based paint removal must be performed in accordance with applicable regulations including 29 CFR 1926 and proper safe work procedures. Alternative methods, such as chemical paint strippers and frozen carbon dioxide blasting may be useful in removing lead based paint systems. All hazardous blast media and other lead paint waste must be properly controlled, collected, stored, and disposed according to the applicable hazardous waste regulations. Refer to 4.4 for appropriate safety procedures.

All unknown existing coating systems should be tested (and any lead-based paint systems found should be identified) before any coating removal is performed in the facility.

Encapsulation of the existing lead paint system may be an appropriate alternative method. This is accomplished by removing only the defective coating and then applying suitable topcoats to the structure.

10.2.5.3 Coat Application

Coating should begin as soon as the surfaces are properly prepared and before formation of any form of corrosion or contamination from atmospheric moisture. Should the prepared surface become dirty or rusty before the first primer coat is applied, the surface should be re-cleaned to the degree initially specified. Additional information may be found in the SSPC *Steel Structures Painting Manual* Volumes 1 and 2.

Typical coating application techniques include the following: (a) brush, (b) roller, (c) spray.

10.2.6 QUALITY CONTROL

All work performed should be subject to inspection by the owner's representative. Any deficient equipment, materials, or techniques should be noted and corrected. Defective coating work or work not conforming to the specifications should be redone. Items that should be included in a quality assurance program include the following:

a. Pre-surface inspection.
b. Ambient conditions monitoring.
c. Surface preparation equipment evaluation.
d. Surface preparation monitoring.
e. Witnessing coating mixing and application.
f. Inspection of coating equipment and techniques.
g. Wet and dry film thickness checks.
h. Evaluation of cleanliness between coats.
i. Holiday testing.
j. Cure and adhesion testing.

10.2.7 SAFETY

All work must be performed in compliance with existing governmental safety and occupational health regulations, such as respiratory protection requirements in 29 CFR 1910 and 29 CFR 1926 and air pollution requirements. Blast cleaning and grinding should be considered hot work and may require permits.

10.2.8 TANKS

The following items should be considered before performance of work on tanks:

a. Depending upon the level of the roof, the area on top of open floating roofs may be considered a confined space area and should be treated accordingly (see 4.2 and 4.4 and API Publication 2026).
b. All floating roof pontoons and areas between the primary and secondary seals must be checked and monitored with a combustible gas indicator. If a flammable vapor-air mixture is found, corrective action must be taken to provide a safe work area (refer to 4.4, 5.2 and API Standard 2015).
c. No blasting or air powered cleaning shall be done on a floating roof unless the vents, pontoons, and the rim seals are in good condition and protected from the effects of the blast.
d. Due to the filling, emptying, or agitation of product in the floating roof type tank, all coating operations should be coordinated with the owner. No blasting or coating shall be allowed when the tank is being filled and for a sufficient period after being filled to minimize potential exposure to vapors (see 5.2).
e. The blasting equipment, including the nozzle, should be electrically bonded to the tank.
f. No blast media should be allowed on an internal floating roof. Any media that does collect on the roof should be removed.
g. All roof openings should be protected to prevent entry of blast media into tank interior during any cleaning operations.
h. The external floating roof drain should be checked to ensure that it is open after all work (blast cleaning and other) has been completed.

10.2.9 LOADING RACK AREA

The loading rack area contains items that are sensitive and must be treated with caution. In particular, the meters, electrical equipment, and any fire detection and fire protection (nozzles) equipment should be protected during cleaning and coating application. Surface preparation techniques must be selected that best fit into the safety constraints of this area (see 4.4 and 9.3).

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10.2.10 PIPING

Piping should be protected from corrosion attack with coatings that are suitable for the environment and piping service. Buried piping should also be cathodically protected (see 10.4).

The atmosphere-to-soil interface may require additional attention due to the corrosion cell formed at this location. All coatings in this area should be properly applied according to manufacturers specifications and inspected periodically to ensure there is no disbondment, shielding, or water entrapment. The coating should extend at least six inches (15.2 cm) above the ground level and adhere well to the pipe.

10.2.11 API COLOR CODE

The use of API product color code (API Recommended Practice 1637) for identifying the product flowing through lines or valves should be considered for petroleum terminals (see 9.13).

10.3 Internal Tank Lining

The use of internal tank linings should be evaluated during the design and maintenance of tanks. Steel storage tanks are often lined to prevent failure due to internal corrosion attack and product compatibility with tank materials. For tanks in petroleum service the complete interior of underground storage tanks is usually lined while typically only the floor and a small portion of the adjacent shell surface of aboveground storage tanks are lined. For additional information see API Recommended Practice 652.

The decision to line a tank interior should include the following considerations:

a. Corrosion prevention.
b. Tank design and operation.
c. Tank condition and history.
d. Product stored; corrosivity or purity requirements.
e. Environmental considerations.
f. Service change.
g. Upset conditions.
h. Federal, state, and local regulations.

10.3.1 LINING SELECTION

Linings for tanks in petroleum service are normally divided into three classes:

a. Thin Film Linings—Thin film linings are classified as a lining less than 20 mils (.051mm). These linings are usually applied to new tanks and to tanks with minimal corrosion. Coal tar epoxy and epoxy phenolics are two common examples of thin film linings.
b. High Build Film Linings—High build film linings are greater than 20 mils (.051mm) thick. These systems may be used on new tanks as well as older tanks with more extensive corrosion.
c. Reinforced Linings—Reinforced linings are usually high build epoxy or polyester resin linings with the addition of fiberglass mat, glass flake, or chopped glass to provide additional strength.

The selection of which lining system to install should include the following considerations:

a. Intended services of the tank. The lining must be compatible and resistant to chemical attack by the stored commodity.
b. The severity of any corrosion attack (particularly pitting corrosion).
c. The degree of tank bottom flexing expected.
d. Resistance of lining to water and product.
e. Resistance to scraping action of metallic shoe seals.

10.3.2 TANK ENTRY

Most tank lining applications and surface preparation are performed in a confined space and require safety precautions for permit required confined space entry. See 4.4, 29 CFR 1910 and API Standard 2015 for additional guidance.

10.3.3 TANK REPAIRS

All repairs to tanks should be completed and inspected per the guidelines of API Standard 653 before the lining is applied (see 6.1). This process includes repairing all holes or large pits in floor or shell, modifications to tank, and weld repair. This procedure will reduce the chances for mechanical damage to the newly installed lining.

10.3.4 SURFACE PREPARATION

Proper surface preparation is critical to ensure satisfactory adhesion and lining performance. Continuous immersion is considered to be severe service for protective coatings. Surface preparation provides for a clean surface with the proper profile or anchor pattern to enable good chemical and mechanical adhesion to the structure. Usually, abrasive blast cleaning to at least a minimum of a near white metal finish (SSPC-SP10 or SSPC-SP5) is required. All burrs, sharp edges, and weld spatter should be ground out to prevent thinning or pinholes in the lining.

10.3.5 DEHUMIDIFICATION

Dehumidification equipment may be used to reduce and control the humidity level inside the tank before and during coating application. The humidity should be at the level recommended by the coating manufacturer to ensure a good, clean, rust free surface and an adequate cure. The use of such equipment may help ensure that the desired blast cleaning standard is maintained.

10.3.6 APPLICATION

The coating material should be mixed, applied, and cured in accordance with the manufacturer's recommendations.

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Adhesion, film integrity, and performance may be adversely affected if the coating is applied incorrectly. Workers should be trained and qualified in coating application, confined space entry, and potential health hazards, and should be provided with and trained in the use of appropriate personal protective equipment (see 4.4 and 4.7).

10.3.7 QUALITY CONTROL

Only qualified inspectors who have a thorough understanding of lining practices should be used. Each step of the process should be inspected, deficiencies noted, and repairs made before proceeding to the next step. Particular attention should be paid to surface preparation, material mixing, film thickness, adhesion, application temperature, hardness, and discontinuities. See 10.2.6 for additional information.

10.4 Cathodic Protection

Metallic structures, such as buried piping, tank bottoms, and containment walls (sheet piling) in contact with moisture in the soil are subject to localized or general corrosion. One way to prevent this corrosion is to force protective current from an external source onto the structure to be protected to counter or overcome any corrosion activity on its surface. The application of direct current electricity from an external source, an anode, onto the protected structure is called cathodic protection. Design, testing, installation, and maintenance of cathodic protection systems should be done by a person knowledgeable in the processes of corrosion control. Cathodic protection has been successfully applied for many years to mitigate corrosion of buried or submerged metallic structures including piping, cables, and tank bottoms (external); and in some cases, to protect the interiors of tanks containing conductive electrolytes. Anodes have also been installed between tank bottoms and detection/containment liners to prevent corrosion of the tank bottom. See API Recommended Practice 651, NACE International Recommended Practice 0169-92, NACE International Recommended Practice 0285-85, NACE International Recommended Practice 0193-93, and NACE International Recommended Practice 0575 for additional information. The selection of cathodic protection systems should be based upon the design objectives, maintenance, operating costs, and appropriate regulatory requirements.

10.4.1 SYSTEM TYPES

There are two fundamental methods of applying cathodic protection: (a) galvanic systems and (b) impressed current systems.

10.4.1.1 Galvanic Systems

Galvanic systems use metals that are more electronegative than the structure to be protected, to produce the protective current. The anodes connected directly to the protected structure are consumed or sacrificed and replaced periodically. Metals commonly used as galvanic or "sacrificial" anodes are magnesium, aluminum, and zinc in ribbon or cast forms. The anodes are generally buried around or under the protected structure in the same electrolyte. Galvanic anode systems have limited capacity and are generally restricted to low current requirements typical of well coated pipelines or small surface area applications.

10.4.1.2 Impressed Current Systems

Impressed current systems have external power sources to force current from the anode to the protected structure. Impressed currents systems usually employ a rectifier that converts AC to DC and provides the desired voltage that forces the protective current to flow from the anode through the soil or electrolyte onto the protected structure. Improper installation could result in reversal of direction of current flow and result in catastrophic failure of the protected structure.

10.4.2 DESIGN OBJECTIVES

The major objectives of a cathodic protection system design should include the following:

a. Protect tank portions and piping in contact with the soil.
b. Minimize stray current effects.
c. Provide flexibility and capacity so that facility changes may be accommodated.
d. Install rectifiers and anodes to minimize effects on operations and maintenance.
e. Provide monitoring of system and protection performance.

10.4.3 DESIGN CONSIDERATIONS

The following items should be considered during the design of a cathodic protection system:

a. Current and voltage requirements.
b. Soil/electrolyte resistivity.
c. Environment temperatures.
d. Impervious tank dike or other containment liners.
e. Foreign structures.
f. Water table.
g. Coatings used.
h. Isolation from the electrical grounding system.
i. Influence of adjacent cathodic protection systems.

10.4.4 CATHODIC PROTECTION CRITERIA

The criteria for determining the effectiveness of the cathodic protection systems are detailed in NACE International Standards RP-0169-92, RP-0285-85, and RP-0193-93. The most commonly used cathodic protection criteria for terminal facilities are the following:

a. -850 millivolt (MV) potential with system energized and voltage drops other than those across the structure-to-elec-

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trolyte boundary considered.
b. 100 MV polarization or depolarization after interruption.
c. -850 MV polarized potential with system momentarily interrupted.

(These potential measurements are made between the structure and a saturated copper-copper sulfate reference electrode contacting soil adjacent to the tank or piping.)

10.4.5 MAINTENANCE AND RECORDS

To function properly and ensure that adequate cathodic protection is applied to control corrosion, the systems should be periodically checked and monitored for protection level. Any needed repairs should be made promptly.

The design, installation, operation, maintenance, and effectiveness of the cathodic protection system should be properly documented by drawings and record keeping. These documents are often required by regulatory agencies to provide evidence of proper operation and maintenance of the system. Typical documentation would include potential measurements, effectiveness of isolating devices, and rectifier operation. The documentation should be retained as long as required by applicable regulations or for as long as the system is in service.

10.4.6 SAFETY

The installation and operation of a cathodic protection system (particularly an impressed current system) requires special safety considerations including the following items:

a. Before a cathodically protected pipe is separated, an electrical bond cable should be installed across the break point to prevent arcing. The rectifier should be disconnected or deenergized (see API Recommended Practice 2003).
b. Excavation should be done carefully to ensure that cathodic protection anodes or cables are not damaged. Where cables are damaged, water proof splice kits may be used to reconnect the severed cables.
c. If a tank valve or line is to be removed and the tank has an impressed current cathodic protection system, the following safety procedure should be performed:

1. Connect a bond wire between the tank and beyond the valve or line to be removed.
2. Turn off the cathodic protection system.
3. Disconnect equipment to be removed and replaced.
4. Turn on cathodic protection system and remove bond wire.

SECTION 11—STRUCTURES, UTILITIES, AND YARD

11.1 Structures

The requirements for the design of new structures and alterations or additions to existing structures, such as administrative buildings, office facilities, warehouses, and garages, are listed in the applicable codes and standards referenced herein or other applicable local regulations.

11.1.1 BUILDING CODES AND STANDARDS

All structures shall be constructed in accordance with the applicable building codes and standards. The requirements for the design of buildings are listed in the BOCA *National Building Code* (NBC), the IBCO *Uniform Building Code* (UBC), the SBCC *Standard Building Code* (SBC), ACI 318, AISC S310, or other locally recognized codes and standards. The requirements for the means of egress and other construction features are found in the above building codes and NFPA 101.

11.1.2 FIRE SAFETY CODES AND STANDARDS

All occupancies shall adhere to the applicable fire and life safety codes. The requirements for general fire safety are listed in BOCA *National Fire Code*, BOCA *Uniform Fire Code*, The ICBO *Standard Fire Code* (SFC), NFPA 101, NFPA 111, 29 CFR 1910, or other applicable local statutes. See Section 5 for Fire Protection.

11.1.3 SAFETY REGULATIONS

All structures shall comply with occupational safety and health codes and standards. The requirements for the design of new facilities and modification to existing facilities are listed in 29 CFR 1910 and 29 CFR 1926 or other applicable state/local code.

11.1.4 ADA REGULATIONS

The applicable requirements for the design of new facilities and alterations or additions to existing facilities as defined by the Americans with Disabilities Act (ADA) are listed in 28 CFR 36.

11.2 Utilities

11.2.1 GENERAL

The design and layout of terminal and tank facilities should include a comprehensive plan for the utility service systems required for the facility operation. These systems may include the following:

a. Electrical.
b. Storm and sanitary sewers.
c. Potable water.
d. Water for fire fighting.

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e. Steam.
f. Air.
g. Natural gas.
h. Communications.

11.2.2 ELECTRICAL SYSTEMS

Electrical systems should be designed in accordance with API Recommended Practice 540 or other applicable codes and should comply with the appropriate electrical area classifications. Area classification drawings should be prepared and maintained for the facility. The requirements for the design and installation of electrical systems and lighting are listed in NFPA 30, NFPA 70 (*National Electrical Code*), API Recommended Practice 500, API Recommended Practice 2003, and 29 CFR 1910, Subpart S. The following is a list of typical electrical systems that should be provided at terminal facilities:

a. Electrical Power Service and Distribution System: Figure 1 portrays in single-line form a typical electrical service distribution system applicable to terminal facilities. Some general design guidelines are as follows:

1. A three phase grounded neutral service is recommended.
2. Service conductors and feeders to motor control centers and motor starter racks should be sized initially for future load increases. For example, an allowance of 25 to 50 percent may be considered.
3. The service disconnecting means should be either a three-pole circuit breaker or fused switch. Ground fault protection should be provided.
4. Depending on the size and configuration of the facility, consideration should be given to installing the metering cabinet, main service disconnect, main panel board, lighting transformer and panel board, and (where feasible) combination motor starters as factory-assembled coordinated packages.

b. Emergency Shutdown System: Such systems, typically consisting of push buttons, relays, alarm lights, and horn, should be designed to shut down all supply pumps to a loading area, such as the tank truck loading rack, tank car loading rack, or the marine loading location. In addition, the following functions should be performed as applicable:

1. Close all automatically controlled valves in supply lines.
2. Close all solenoid-operated valves at the loading rack.
3. Shut down incinerator for vapor disposal system.
4. Open all motor-operated terminal entrance and exit gates.
5. Sound alarm and notifications to pipeline.

See 9.9 for additional information.

c. Tank Trucks, Tank Car, or Marine Vessel Loading Electrical Systems—Provide the following as applicable:

1. Start-stop control of loading pump motors.
2. Grounding and bonding of loading racks and tank trucks, bonding of loading lines, and stray current protection for loading spur railroad tracks.
3. Loading control and metering system.
4. Tank truck high level shut off system.

d. Card lock or Key lock System—These are typically provided at truck loading installations to provide security and data acquisition capability. Coded cards or keys are used for access to and control of terminal functions.

e. Vapor Collection and Disposal System—Provide for main power, control power, annunciator, and emergency shutdown.

f. High Level Tank Alarms—Provide as required by NFPA 30 and as described in API Recommended Practice 2350.

g. Auxiliary Facility Systems—Provide electrical service for own-use fueling, pump back, meter prover, and truck engine heaters. The service should include pump controls, interlocks, and disconnects as required.

h. Lighting—Sufficient lighting and proper electrical classification should be provided for loading racks, yard and tank farm areas, gate card reader, own-use fueling, pump back and meter prover facilities, loading pumps, motor starter racks, office, garage and warehouse lighting. One source of information on this subject is 29 CFR 1910. The following general guidelines also apply:

1. Illumination of outdoor areas may be achieved with various types of high intensity discharge lamps, such as mercury vapor or high pressure sodium lighting. Generally, high pressure sodium lighting is recommended because it provides high lumen output per watt.
2. Application of mercury vapor and high pressure sodium lamps at low temperatures should be referred to the manufacturer for special consideration.
3. On energizing, high intensity discharge lamps, such as mercury vapor and high pressure sodium lamps, require time to reach full lumen output. Areas that require immediate return of lighting after power dips or outages should be provided with interspersed incandescent lighting fixtures. The use of instant restrike lighting eliminates the need for interspersed incandescent lights.
4. Photoelectric cell control should be considered where automatic switching of yard and rack lighting is required.
5. Lighting fixtures installed in Class I, Division 1 and 2, Group D locations should conform with the requirements of NFPA 70.
6. Explosion proof lighting installed in Class I locations should be maintained in good condition.

See 9.2 and 11.3 for additional information.

i. Motor Starters and Motors—Motor starters should have a circuit breaker, a temperature-compensated overload relay, and except where a common control power transformer has been approved, a fused control power transformer. The control power transformer should be sized to supply the additional burden of control relays, indicating lamps, or

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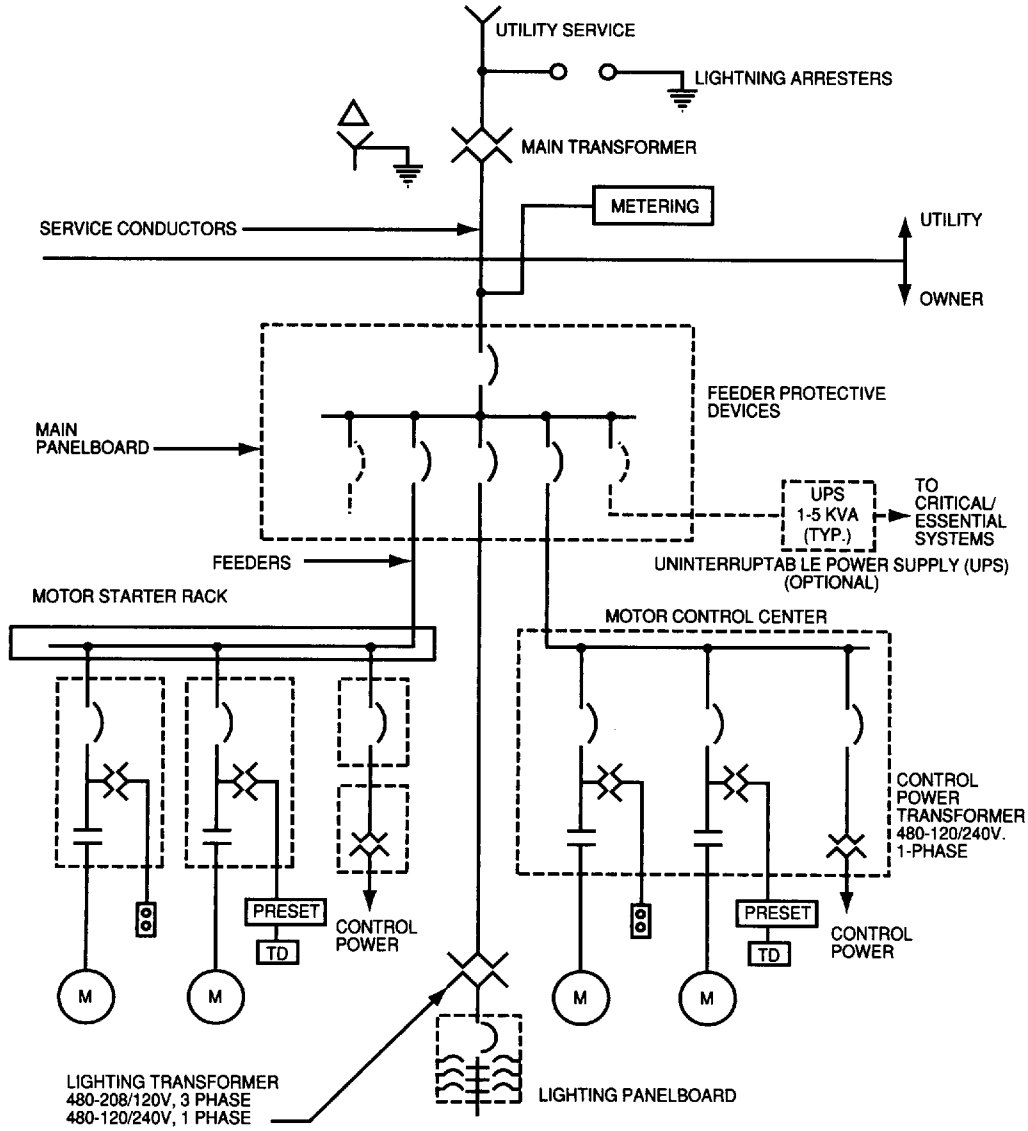


Figure 1—Typical Electric Service and Distribution System Terminals and Bulk Plants

solenoids that may be connected in the control system. Motors, in general, should be supplied with totally enclosed, fan-cooled (TEFC) enclosures. Explosion proof motors are required if they are installed in Class I, Division 1 locations. Motors should comply with the requirements of NFPA 70 and API Recommended Practice 500.

j. Conduit and Wiring—The following guidelines should be considered for the design and installation of conduit and wiring:

1. Above grade conduit should be rigid aluminum or galvanized steel. Below grade conduit should be rigid galvanized steel.
2. Conduit may be routed above grade on pipe ways, sleepers, or other structures. In the absence of structural supports, conduits should be installed below grade. To minimize corrosion, below grade conduits should be completely covered with a corrosion protective coating (see 10.2). Where additional mechanical protection is required for safety, a red concrete slab (or block) should be installed over the conduit runs. Alternatively, the conduits may be encased in concrete.
3. Wire and cable should be supplied with copper conductors.
4. In Class I, Division 2 and all nonclassified outdoor locations, conduit connections to motors, valves, set stops, and other equipment that may move or vibrate

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should be made with flexible conduit that meets the requirements of the electrical classification.

5. All conductors should be identified.
6. Where required, fire protected cable/conduit shall be used.

k. Terminal Automation Systems—Consideration should be given to providing a computer based system that includes the following typical functions:

1. Records and reports product transfer volume information based on meter and user coded inputs, usually at locations with truck loading racks.
2. Receives and transmits accounting information (pricing and product inventories) between the terminal and a central office.
3. Monitors and records terminal access and egress information based on gate security inputs, such as card lock or key lock systems.
4. Where equipped, monitors tank inventory and pipeline receipts.

l. Uninterruptible Power Supply (UPS)—Consideration should be given to providing a UPS system for terminal automation systems, computers used to control loading systems, and other systems that warrant a back-up power supply.

11.2.3 WASTE WATER COLLECTION AND TREATMENT SYSTEMS

The design of waste water collection and treatment systems for a petroleum terminal facility should be based on the segregation of contaminated sources of waste water from noncontaminated sources and the elimination or minimizing of effluent discharge to adjacent property and waterways. This design procedure will reduce treatment cost and protect water quality. For further information, also see API Publication 4602.

Separate systems should be considered for sanitary sewer, for noncontact storm water, and for petroleum contact waste water including contact storm water. Further segregation may be appropriate based on contaminant concentration or stream source.

Existing facilities may have only two separate systems: one for sanitary waste and a combined system for noncontact storm water and petroleum contact waste water. In some cases a facility may be permitted to discharge a portion of its waste water to a Publicly Owned Treatment Works (POTW) rather than to a surface receiving water under the NPDES permits.

Refer to Figure 2 for a simplified flow diagram for a typical terminal showing segregated waste water streams.

The following guidelines should be considered in the design of waste water collection and treatment systems.

11.2.3.1 Sanitary Waste Water System

a. The sanitary system includes waste from all toilets, showers, and washbasins. Sanitary waste should be discharged to the local municipal sewer system. Where municipal sewers are not available, sanitary waste should be discharged to an on site treatment/disposal system, such as a septic system or a package treatment unit.

b. The size of the sanitary system is governed by the total number of people normally using the facility. Per capita use of water is usually 50 US gpd (0.2m3/day). For certain types of occupancy, local authorities may specify minimum quantities per person.

c. Where public sanitary sewers are not available, septic systems are the preferred design. Designs of septic systems should be developed per local code requirements and approved by local Public Health Authorities where applicable.

d. Where soil conditions do not permit discharge of effluent to a leaching field, or where local regulations prohibit septic systems, a package treatment plant should be considered. Small units, available in various capacities, consist of an equalization section, an aeration section in which aerobic bacterial action is maintained by mechanical agitation or diffuser aerators, a settling section, and a disinfection section. An aboveground steel unit is the recommended design.

11.2.3.2 Noncontact Storm Water System

a. The noncontact storm water system includes storm water runoff from nonoily areas, such as building roofs, loading rack roofs, sidewalks, roadways, uncontaminated yard parking and storage areas, and grassy "green belt" areas. Paved areas at truck loading operations in bulk plant facilities, including storage areas and material handling areas where drips and spills may occur, should not be considered "clean" yard parking.

b. Noncontact storm water should be monitored and if satisfactory, discharged directly to a municipal storm water sewer system or natural drainage area, such as an available water way, provided that any applicable NPDES or other local regulatory specifications are met.

c. Where tank dike areas are determined to be clean, treatment of the dike area storm water may not be necessary depending on local regulations. Before storm water drainage, diked areas should be visually inspected for contaminants. A means of diverting storm water to the oil-contaminated waste water system should be provided.

d. Factors affecting the design of an adequate storm water system at a specific location include facility layout, applicable governmental regulations, local rainfall, and topography. Refer to 3.3.2 for discussion of items impacting noncontact storm water systems.

e. The rate at which storm water enters the system depends on storm intensity, area drained, topography, type of surface, and time of concentration. The time of concentration is the time required for storm water to flow from the farthest point of the catchment area to the treatment system component being sized. The design runoff flow rate entering the sewer

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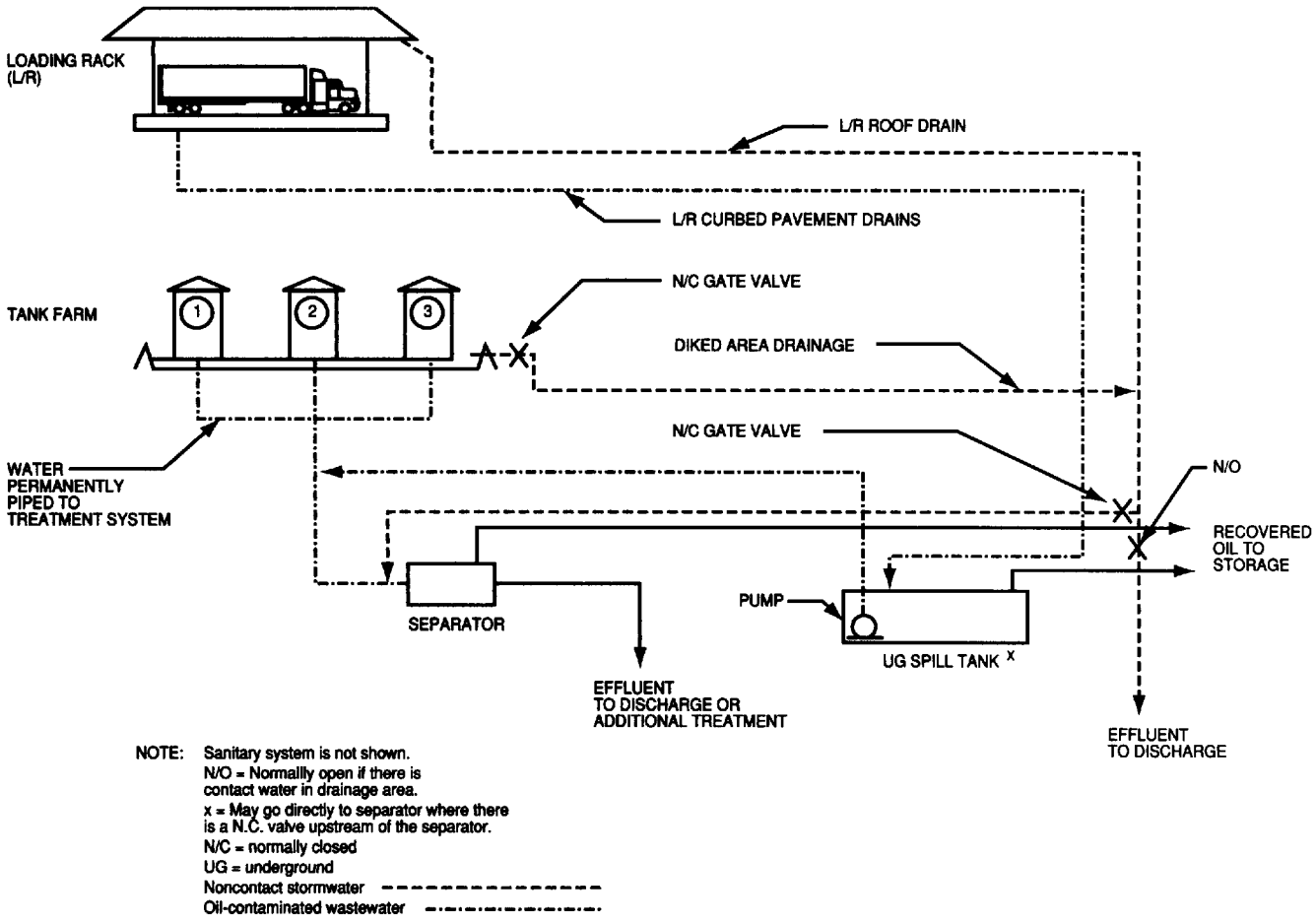


Figure 2—Segregated Waste Stream in a Typical Terminal

should be based on the greater of the 10-year frequency, one-hour duration storm intensity, or designed fire water volumes. Necessary rainfall data may be obtained from the local weather bureau or similar government offices. Storm water impoundment volumes should be based on the 10-year frequency, 24-hour duration storm intensity with consideration for fire water volumes.

11.2.3.3 Petroleum Contact Waste Water System in Petroleum Terminals

Sources of petroleum contact waste water may include:

a. Loading areas.
b. Tank water drawoffs.
c. Recovered groundwater.
d. Truck washing.
e. Drainage from loading rack spill containment systems.
f. Other areas: warehouse and garage washings, drips from curbed pump pads and process equipment, vapor recovery unit pads, and laboratory wash water.
g. Diked areas: If water from diked areas is contaminated, it should be discharged into the petroleum contact waste water system.
h. Hydrostatic test water.

11.2.3.4 Waste Water Treatment

Treatment of petroleum contact waste water or both non-contact storm water and petroleum contact waste water may be necessary depending on local regulatory requirements. See 3.3.2 for further information on waste water treatment.

11.2.4 PIPING BASED UTILITY SYSTEMS

11.2.4.1 The design and installation of piping based utility systems, such as potable water, fire water, steam, air, and natural gas should be done in accordance with the applicable piping engineering practices, codes, and standards, such as the following:

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a. ASME B31.3.
b. AWWA C100.
c. NFPA 24.

11.2.4.2 Utility piping should be clearly marked with appropriate colors and identification. Also, appropriate end fittings should be installed to prevent connection errors.

11.2.5 COMMUNICATION, SECURITY, AND CONTROL SYSTEMS

The design of communications, security, and control systems, such as telephone, card locks, security cameras, automated equipment and intercoms, may be performed as part of the services supplied by the vendors of the communications equipment. These systems should be installed so that service is maintained during emergency situations, such as a fire.

11.3 Yard

11.3.1 YARD FUNCTIONS

The yard layout for new facilities should be arranged so that all truck operations may be accomplished in a sequential and efficient manner from the truck entrance to the truck exit. Primary yard functions or requirements to be considered are as follows:

a. Ingress and egress from the street.
b. Traffic pattern and adherence to one-way traffic.
c. Location of the office and garage facilities. (The dispatcher's or supervisor's office should have a clear view of the loading rack if not monitored by remote cameras.)
d. Driver instructions for loading and delivery.
e. Truck loading.
f. Recording of loading information.
g. Driver administrative transactions.
h. Vehicle staging or vehicle inspection before entering the loading rack.
i. Vehicle parking during shift changes.
j. Temporary truck parking while drivers are getting instructions.
k. Employee, visitor, and contractor parking.
l. Fire fighting, fire prevention, and protection.
m. Eyewash and safety shower.
n. Driver rest room and locker room as required.
o. A 24-hour telephone or other means of emergency notification.

Secondary yard functions that should be considered include the following:

a. Telephone.
b. Loading meter proving.
c. Pump back for product return and for flushing and draining trucks for switch loading.
d. Drive-through truck wash or garage with wash bay.
e. Own-use island for truck fueling.
f. Compressed air for truck tire inflation.
g. Hot-start facilities for warming engines during cold weather.
h. Flat concrete pad for parking trailers when connecting and disconnecting trailers from tractors.

11.3.2 COMMUNICATIONS

For transmittal of loading instructions and records between truck drivers and dispatch office, the following alternatives should be considered:

a. Elevate the dispatch office and install a pass-through window so that the dispatcher in the office may hand papers to a driver in a truck.
b. Install an intercom or pneumatic tube or both between the dispatch office, truck entrance and exit station or stations, and the loading rack.
c. Have the driver enter the office for all dispatched transactions and instructions.
d. Install a small building or shed on loading islands for key or card input and instructions. Location of this building should be in accordance with local codes or safety standards.
e. Install a small building or shed away from the loading rack for bill of lading (BOL) tickets so others may load while the driver picks up the BOL.

11.3.3 YARD ARRANGEMENTS AND DIMENSIONS

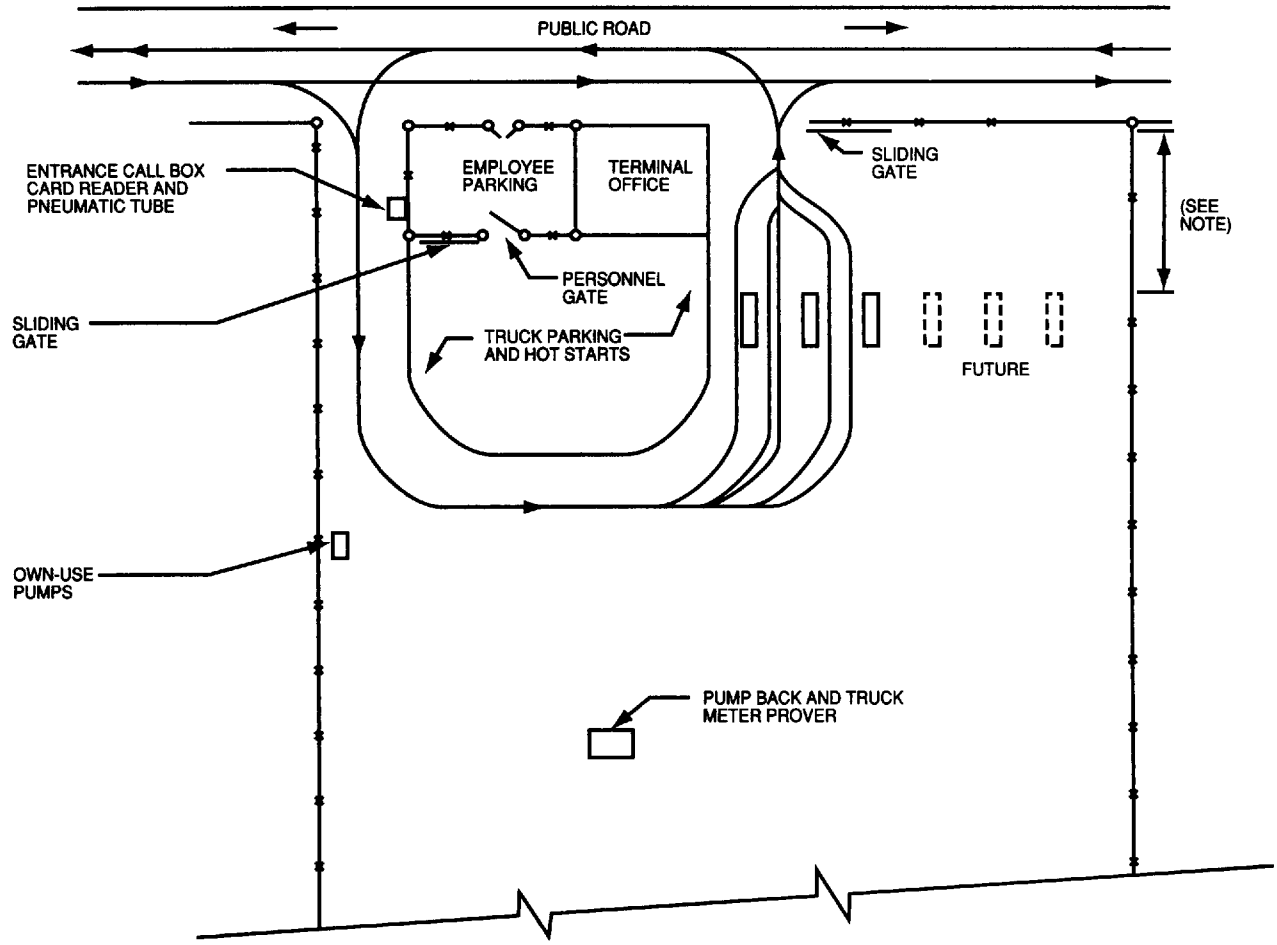
11.3.3.1 Example layouts for tank truck loading yards are shown in Figures 3 and 4. Figure 3 illustrates an installation without a garage. Figure 4 illustrates an installation with a garage. The loading rack should be located so that there is a minimum of truck maneuvering within the yard area, particularly after loading. It is usually more efficient to locate the garage on the entrance side of the terminal away from the loading rack so that the trucks may be serviced before loading. Left turns for trucks are preferred to right turns for better visibility.

11.3.3.2 Entrance and exit roads should be sufficiently wide to permit a moving truck to pass a parked truck, or if practical, provide a minimum width of 20 feet (6.1 meters). If the yard gate is near the street curb line, the road and gate should be 30 feet (9.1 meters) wide, if practical, to allow extra space for a vehicle turning in from the street. At terminals where the driver must stop and open the gate, the entrance gate should be recessed from the road a distance equal to the longest vehicle.

11.3.3.3 An example of typical yard dimensions for movement to and from the loading rack are shown in Figure 5. These dimensions are based on trucks that are 8-1/2 feet wide (2.5 meters), require a turning radius of 60 feet (18.3 meters), have 55 foot long (16.8 meters) semitrailers, and 70 foot long (21.3 meters) tank trucks with full trailers. Recom-

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NOTE: Allow enough space between loading racks and gate, at least two truck lengths, for trucks to be parked for driver settlement

Figure 3—Tank Truck Loading Yard Layout Without Garage

mended yard widths are designed to allow a truck to turn 90° and obtain a straight path by the time it reaches the loading position. In locations with limited area, the minimum yard width may be used; however, truck maneuvering is more difficult and parked vehicles may cause congestion. The yard dimensions for bottom loading racks shown in Figure 5 allow an extra 15 feet (4.6 meters) of length at the rack exit to ensure truck clearance to the safety gate.

11.3.3.4 Areas for own-use fueling of trucks as well as pump-off (pump back) and drive-through wash stands should usually be located on the loading rack approach so that the trucks have a minimum deviation from the normal route through the yard to the rack.

11.3.4 YARD DRAINAGE

11.3.4.1 The yard should be sloped to drain rainwater away from the concrete mat at the loading rack, truck fueling, truck scales, and pump back areas. The grade in the yard paving should be enough to ensure that irregularities in the paving will drain without the formation of puddles. Proper grade is especially important in freezing climates where ice spots become a hazard (see 4.2). Where feasible, the typical slope should be 1 percent.

11.3.4.2 The discharge and disposal of yard drainage water will depend on local regulations and whether the storm water drainage is determined to be clean or contaminated (see 3.4.2). Provisions for sampling and observation of storm water drainage should be included in the design. For disposal of uncontaminated storm water from small areas, local policy may permit the water to simply drain into the street. In large areas, where sufficient grade may not be provided for natural runoff, catch basins, drain piping, or

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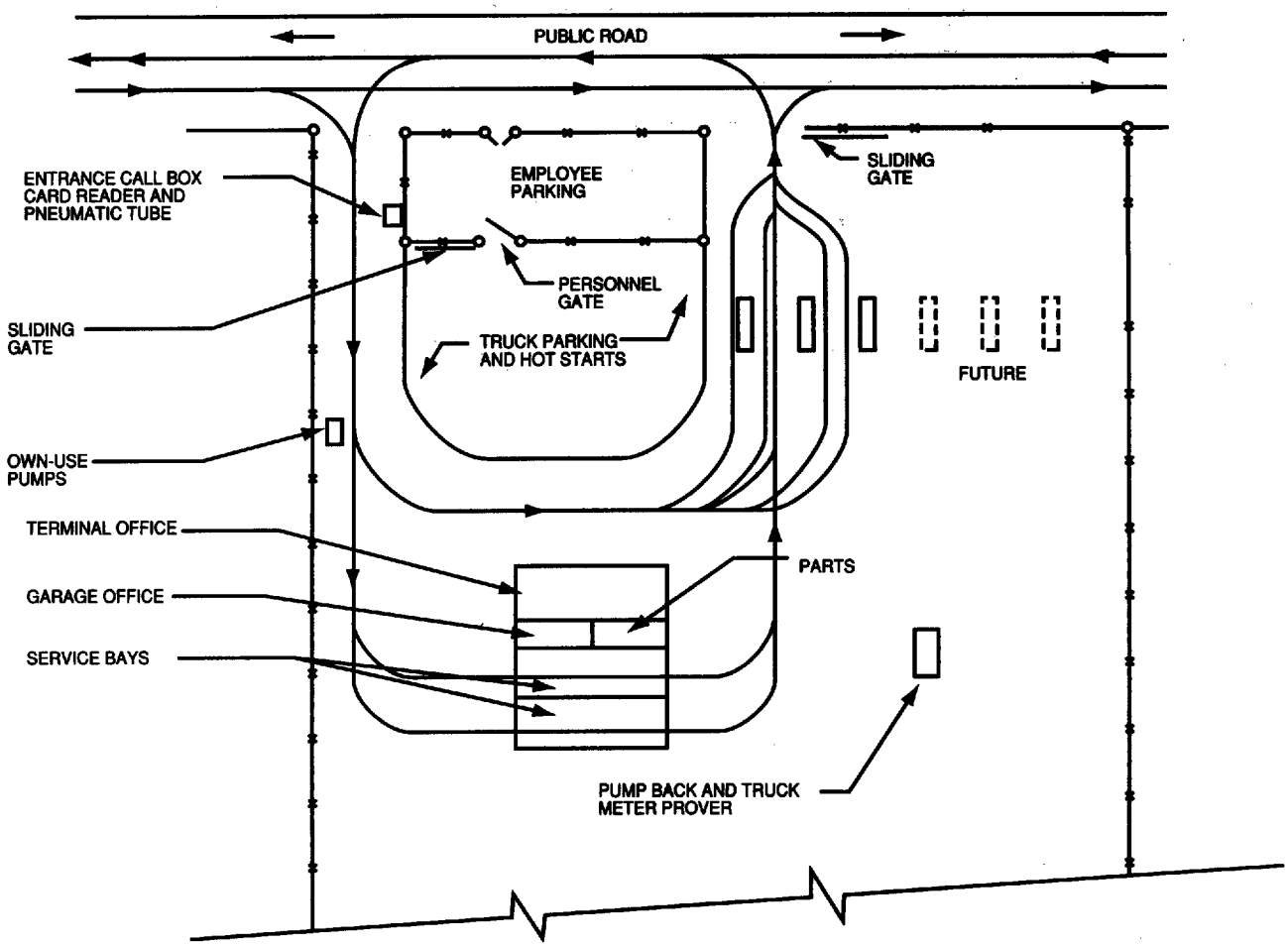


Figure 4—Tank Truck Loading Yard Layout With Garage

pumps may need to be installed for drainage disposal. Discharge and disposal of storm water are covered in 3.3.2.

11.3.5 YARD PAVING

The yard paving is usually asphalt with concrete required for loading areas. Concrete paving may be used at the exit from the loading rack to provide a more durable surface. Asphalt pavement is subject to scuffing when the rear tandem wheels of the tractor and semitrailer, that are not articulated, are dragged across the surface as the truck is turning. Where scuffing of asphalt may present a problem, concrete should be considered. Also note that asphalt may be subject to degradation upon contact with petroleum products.

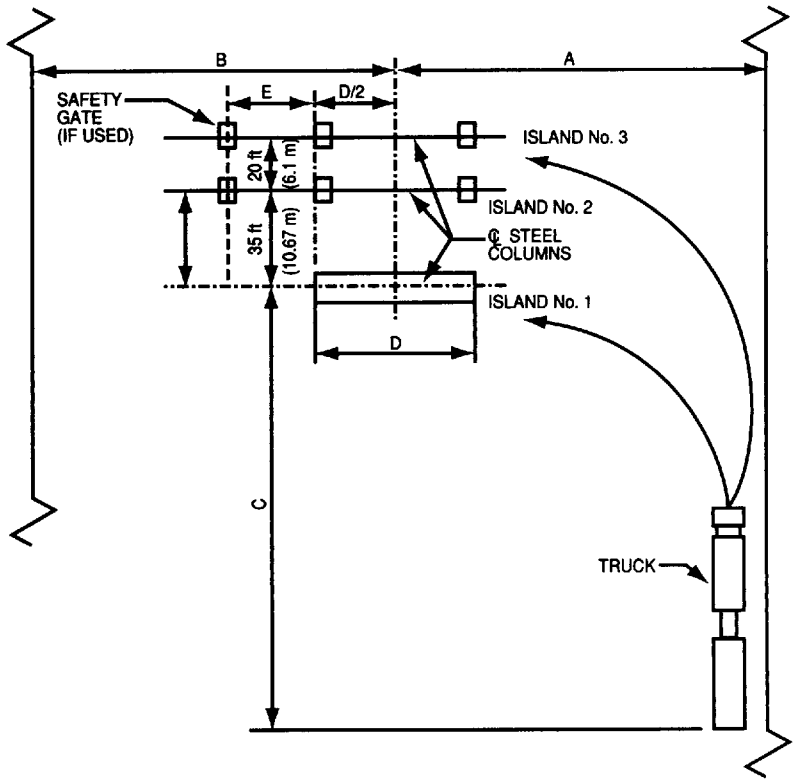
11.3.6 FENCING AND SECURITY

11.3.6.1 Fencing around the entire site should be employed to maintain facility security and prevent product loss and vandalism. The location of the fence line in relation to the property line shall conform to local code requirements.

11.3.6.2 A review of the security requirements for the terminal facility should be included during the design consideration. The review should include physical security, disaster recovery planning, security incidents, security awareness, and security technology.

11.3.6.3 Perimeter lighting may be provided to illuminate fences, access and egress gates, employee and truck parking, tank farm and manifold areas, loading racks, and marine docks. Refer to the *IES Lighting Handbook* or other applicable state or local code.

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Trucks	Trucks Length	Yard Width				Yard Entrance	Island Lengths	Safety Gate
		Minimum		Desirable			Top or Bottom Loading	Bottom Loading
		A	B	A	B	C	D	E*
Semitrailers (Supertruck)	55 ft (17 m) 60 ft (18 m)	100 ft (30 m)	115 ft (35 m)	120 ft (37 m)	135 ft (41 m)	130 ft (40 m)	35 ft (11 m)	32 ft (10 m)
Semitrailers & Trucks with Full Trailers	55 ft (17 m) 70 ft (21 m)	105 ft (32 m)	120 ft (37 m)	125 ft (38 m)	140 ft (43 m)	140 ft (43 m)	48 ft (15 m)	28 ft (8.5 m)

Island No. 1 - Top Loading both sides.
Island No. 2 - Bottom Loading left side only.
*This dimension should be checked in the field because of the different truck types and location of connections.

NOTE: Dimension B allows extra 15 ft (4.6 m) for truck to clear safety gate.

Figure 5—Yard Dimensions for Trucks (See Table Above)

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SECTION 12—REMOVALS AND DECOMMISSIONING OF FACILITY

12.1 General

12.1.1 The guidelines in this section apply to the removal or decommissioning of facilities including the following: buildings, tanks, pumps, piping, foundations, steel structures, electrical equipment, or any other equipment that is part of a terminal facility.

12.1.2 Removal and decommissioning work requires proper planning, control, and close-out procedures. The most important considerations for all phases of this type of work should be to ensure the safety of all involved.

12.2 Site Control and Protection

12.2.1 A representative of the facility owner, knowledgeable in its operation and familiar with this type of equipment, should be present during the removal work to assist with the safe and proper execution and appropriate documentation of the work.

12.2.2 Adequate control measures should be provided to protect personnel, equipment, surrounding environment, and adjacent property from injury or damage from falling debris, dust, vibration, product releases, or other hazards that may occur during the work. See 4.4 for a further discussion of safe work practices. All work shall be performed in accordance with the applicable federal, state, and local regulations.

12.3 Preparations

12.3.1 Tests should be conducted to determine the presence of hazardous substances or materials that would require special handling precautions during removal. These hazards would include asbestos and man-made fibers (found in some insulating materials), lead (found in tanks that have stored leaded gasoline or in some paints and primers), and contaminated soil (may be found in areas in proximity to equipment, such as tanks, loading and unloading areas, and pumps).

12.3.2 All safety and operational clearances and permits, including any jurisdictional permits shall be obtained. This includes preparation of manifests for any hazardous wastes that have been identified by the tests conducted according to 12.3.1.

12.3.3 The equipment to be removed shall be positively isolated and de-energized from all active equipment and interconnecting facilities, piping, and wiring (see 4.4).

12.3.4 All product, gases, and contaminants shall be removed from the equipment to be removed. Disposal of equipment, contaminants, and other contents shall be in accordance with operational and jurisdictional requirements. This includes such work as cleaning tanks and emptying piping systems by flushing or mechanical means. One source of information is API Publication 1638 for guidelines on the disposal of materials. Another reference is API Standard 2015 for guidelines on the cleaning of storage tanks.

12.4 Execution

12.4.1 Removal or idling work shall be performed in accordance with safe work practices outlined in 4.4.

12.4.2 Consideration should be given to removing underground piping that is being de-commissioned. However, if local regulations allow underground piping to be idled in place, adequate safeguards should be provided, such as evacuating the piping, and plugging and capping the ends of piping to prevent potential contamination.

12.4.3 If aboveground storage tanks are to be idled in place, adequate safeguards against unauthorized entry, such as providing bolted covers on manways and nozzles should be provided. A means to prevent flotation of the empty tanks by an accumulation of surface storm water, such as providing adequate drainage around the tank, should be provided. In areas subject to heavy rains or snow, tank external floating roofs should be supported by cribbing or other methods to prevent damage to the roof. Additionally, tanks in areas subject to extreme wind conditions, such as hurricanes, should be protected by anchoring or reinforcing as applicable.

12.4.4 If equipment such as storage tanks or underground piping may be recommissioned in the future, then consideration should be given to corrosion protection systems, such as cathodic protection systems or protective coatings. See Section 10 of this standard for further information on this subject.

12.5 Site Assessment and Remediation

As part of equipment removal work, tests should be conducted to assess site contamination. Refer to API Publication 1628 and API Publication 1629 for guidelines on site assessment and remediation.

12.6 Closeout and Cleanup

12.6.1 All debris and waste are to be disposed of in accordance with applicable regulations, recognizing the potential future liabilities associated with improper disposal. See API Recommended Practice 1638 for guidance on tank bottoms disposal and API Publication 2202 for guidelines on disposing of steel scrap from leaded gasoline storage tanks.

12.6.2 At project completion, temporary structures and safeguards should be removed and remaining buildings and equipment restored to a clean and usable condition.

12.6.3 The finished grade should be smooth and continuous with the surrounding finished grades. New grades should be protected against erosion.

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